STAGE: A Full-Screenplay Benchmark for Reasoning over Evolving Stories

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Abstract

Movie screenplays are a demanding testbed for long-form narrative understanding, as characters' goals, beliefs, knowledge, and relationships evolve continuously across scenes. However, existing benchmarks primarily evaluate isolated facts from the completed screenplay, leaving unassessed whether models can track the evolving state of characters as the story unfolds. We introduce STAGE, a benchmark over 151 English and Chinese full-length screenplays, built on a provenance-linked narrative backbone that recovers the state and epistemic access of each character at every point along its timeline. Three tasks derived from the backbone jointly probe whether models can maintain, explain, and act on evolving narrative state: *Character Development Tracking* updates a focal character's state between checkpoints, *Cross-Scene Narrative Evolution Reasoning* targets cross-scene state transitions, and *In-Script Character Role-Playing* requires responses bounded by the character's state and knowledge at a specified point. We identify three failure modes of current LLMs: silent forgetting under recursive state updating, limited cross-scene reasoning even when all relevant evidence is supplied, and a trade-off in role-playing where stylistic character fidelity and screenplay-grounded memory faithfulness are optimized by different memory-access strategies. STAGE thus provides a unified framework for diagnosing how current models fail to track, reason about, and enact story evolution. Benchmark assets and evaluation code are available at `https://github.com/roytian1992/STAGE_v0`.

1 Introduction

A screenplay presents an ordered sequence of state changes. As the script proceeds, a character's goals are revised, beliefs are corrected, relationships are reconfigured, and constraints are imposed and lifted. Understanding such a text requires a time-indexed representation of the story, covering the currently valid state, the cause of each change, the characters with access to it, and the elements that persist unchanged (Graesser and Franklin, 1990; Graesser et al., 1994; Riedl and Young, 2010). A question about an intermediate stage of the story therefore calls for checkpoint-specific state reconstruction. Retrieval over the completed script remains insufficient for this purpose, even for long-context LLMs (Liu et al., 2024). Such state reconstruction is broadly useful in settings with an accumulating interaction history (e.g., long-horizon agents (Park et al., 2023; Xu et al., 2025; Tan et al., 2025)), and the explicit scene order and textually realized state changes of screenplays make them a natural starting point for studying it.

Evolving narratives introduce two asymmetries that static document comprehension often leaves implicit. First, evidence accumulates, but state validity does not: an earlier goal, belief, relationship, or constraint may be revised, resolved, or superseded. Second, narrative truth and character knowledge diverge. An event can alter the story world without being observed by the focal character, and a later revelation cannot license an earlier action. Evaluation that ignores these temporal and epistemic boundaries may reward a factually plausible answer even when it is invalid at the queried point.

Existing screenplay and long-form narrative benchmarks primarily emphasize recognition, summarization, conditioned generation, story QA, or general long-context reading (Zheng et al., 2025a; Gorinski and Lapata, 2015; Papalampidi et al., 2020; Saxena and Keller, 2024; Chen et al., 2024b; Mirowski et al., 2023; Zheng et al., 2025b). Related long-form benchmarks broaden document-level coverage (Kočiský et al., 2018; Bai et al., 2024; Wang et al., 2025; Hamilton et al., 2026). However, they generally treat the completed story as a static source or assess tasks independently, leaving open whether models can track accumulating state, explain changes at the correct time, and

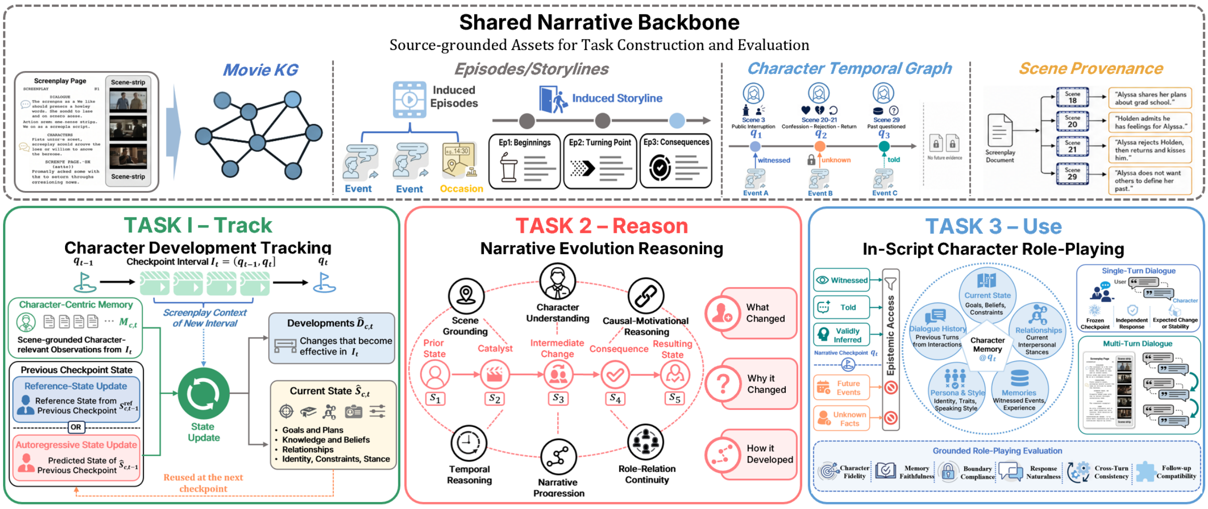

Figure 1: Overview of STAGE. Ordered screenplay scenes support a source-grounded representation with an objective narrative backbone and character-relative temporal views. These construction assets remain hidden from evaluated models and define three aligned tasks: **Character Development Tracking** measures successive state updates and rollout stability, **Cross-Scene Narrative Evolution Reasoning** evaluates explanations of state change, and **In-Script Character Role-Playing** tests checkpoint-valid generation.

act within the knowledge available at that point. This is also a cross-use alignment problem: a model may recover the right scene for a question yet fail to carry its consequence into a later state update, or imitate a character fluently while exceeding that character's acquired knowledge. Aligning tracking, reasoning, and generation to the same ordered evidence enables direct evaluation of all three capabilities.

To address this gap, we introduce STAGE (Screenplay Text, Agents, Graphs & Evaluation), a full-screenplay benchmark covering 151 English and Chinese movies. Its three tasks share a temporal axis: **Character Development Tracking** updates a focal character between checkpoints and measures recursive error accumulation; **Cross-Scene Narrative Evolution Reasoning** tests evidence-grounded reasoning about change, timing, knowledge, delayed effects, and persistence; and **In-Script Character Role-Playing (ICRP)** requires natural responses bounded by the character's checkpoint-valid state and knowledge. Together, they test whether models can *track*, *reason about*, and *use* story evolution. To define these tasks, we need each character's state and knowledge at every checkpoint, which remain latent in screenplay text. We therefore construct a narrative backbone for each film, comprising a source-grounded objective narrative hierarchy and a character-relative temporal graph, and derive the tasks from it. By recording evidence, checkpoints, state validity, and epistemic access once per film, the backbone makes annotation tractable at full-screenplay scale and keeps the three tasks grounded in the same source, while remaining hidden from evaluated models.

Our evaluation identifies three complementary failure modes: recursive updates progressively omit prior developments; cross-scene answers remain unstable even when retrieval covers all gold evidence; and role-playing can remain fluent while failing to use checkpoint-valid memory faithfully. Collectively, our contributions are:

- **An aligned evolving-story benchmark** spanning tracking, cross-scene reasoning, and checkpoint-bounded generation over 151 films.
- **Source-grounded construction** that separates objective narrative structure from character-relative state and knowledge.
- **A systematic capability analysis** showing recursive state error, unstable cross-scene evidence use, and fluent but weakly grounded role-playing.

2 Related Work

Narrative structure and evolving state. Computational narrative theory models comprehension through goals, causality, plans, and character intentionality across events (Graesser and Franklin, 1990; Graesser et al., 1994; Riedl and Young, 2010). It also distinguishes event ordering from higher-level narrative structure: Piper et al. (2021) organize the latter as an event–scene–narrative-level–plotline–plot hierarchy. Event-centric graphs make parts of this structure explicit (Yan and Tang, 2023), while consistency studies show that language models can vary across contexts even when the underlying facts are unchanged (Novikova et al., 2025). STAGE operationalizes this coarse-to-fine principle for screenplays and adds a character-relative temporal layer. This supports checkpointed evaluation that separates objective events from character access, and current states from the developments that produced them.

Long-form benchmarks and retrieval. NarrativeQA, BookQA, LongBench, NovelQA, and TLDM evaluate retrieval and document-level comprehension across books, scripts, and mixed long documents (Kočiský et al., 2018; Angelidis et al., 2019; Bai et al., 2024; Wang et al., 2025; Hamilton et al., 2026). Screenplay resources emphasize analysis, summarization, or conditioned generation (Gorinski and Lapata, 2015; Papalampidi et al., 2020; Saxena and Keller, 2024; Chen et al., 2024b). These tasks establish the difficulty of long narrative context, but generally evaluate the completed document through one output interface. STAGE aligns three interfaces to the same temporal backbone. Its QA comparison also contrasts flat retrieval (Robertson and Zaragoza, 2009; Lewis et al., 2020; Izacard and Grave, 2021), hierarchical access such as PageIndex (Zhang et al., 2025), and graph-informed access (Edge et al., 2024; Guo et al., 2025a; Yang et al., 2025b; Gantt et al., 2024; Luo et al., 2025).

Character role-playing and memory. Persona-conditioned dialogue, fictional role adoption, character profiling, and long-term agent memory are studied by Persona-Chat, CharacterGPT, PersonaEval, TVShowGuess, PersoNet, and LoCoMo (Zhang et al., 2018; Shanahan et al., 2023; Park et al., 2025; Zhou et al., 2025; Sang et al., 2022; Yu et al., 2023; Maharana et al., 2024). Most systems condition behavior on a static profile, a known role, or dialogue history. ICRP fixes a screenplay checkpoint, so persona expression must remain compatible with both evolving state and the information available at that point.

LLM-assisted construction and evaluation. Open-ended benchmark construction and scoring increasingly use LLMs under explicit quality controls (Li et al., 2025; Bai et al., 2023; Gu et al., 2025; Zheng et al., 2023). Reference-conditioned and factuality methods reduce unconstrained preference judgments by grounding decisions in claims or evidence (Badshah and Sajjad, 2025; Min et al., 2023; Liu et al., 2025; Thorne et al., 2018; Fan et al., 2020). STAGE follows this evidence-first design: reference answers, source scenes, contradiction targets, and knowledge-boundary scaffolds are retained for evaluation, and judge reliability is calibrated against blinded human annotations.

3 The STAGE Framework

3.1 Dataset

3.1.1 Data collection and statistics

STAGE comprises 151 feature-length movie screenplays (109 English and 42 Chinese) collected from publicly accessible screenplay repositories and script documents. We retain scripts with recoverable scene boundaries, dialogue, and action descriptions, excluding summaries, transcripts, and fan-written adaptations. After normalizing scene structure and OCR artifacts, we create the primary original-identity corpus. The public release also includes pseudonymized counterparts that differ only in approved work-specific entity forms; scene order, content structure, and task semantics remain unchanged. The scripts average about 30K words and 151 scenes; detailed distributions by language, genre, release year, length, and scene count appear in Appendix A.

3.1.2 Narrative backbone construction

The STAGE narrative backbone is a provenance-linked, multilevel representation that links screenplay evidence to movie-level narrative structure and to each character's state at a given story point. It adapts the event–scene–plotline progression in computational narratology (Piper et al., 2021); Figure 2 summarizes six provenance-preserving construction stages.

The first three stages normalize ordered scenes, extract source-bearing Events, Interactions, and Occasions, and canonicalize entities and typed relations (Figure 2a–c). Ordered scenes are segmented at semantic boundaries with complete source coverage. An *Event* records a bounded action or change, an *Interaction* a process involving multiple characters, and an *Occasion* a reusable activity or situation. Each unit retains its supporting text spans, while aliases and typed relations are canonicalized in a shared entity registry.

The next two stages group scene-local units into Episodes and connect Episodes across scenes as

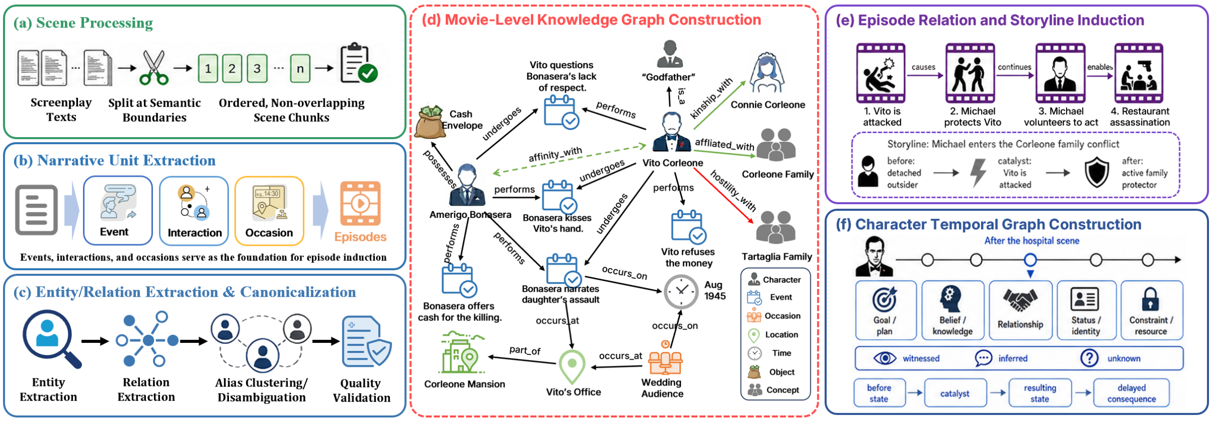

Figure 2: Six-stage construction of the STAGE narrative backbone, illustrated with *The Godfather*: (a) scene processing; (b) narrative-unit extraction; (c) entity and relation canonicalization; (d) movie-level KG construction; (e) Episode-relation and Storyline induction; and (f) character temporal-graph construction. Every retained asset preserves its supporting screenplay scenes.

Storylines (Figure 2d–e). Each Event or Interaction belongs to a primary Episode that captures its setup, development, and outcome. Storylines then connect these Episodes through an explicit prior state, catalyst, resulting state, and scene provenance.

The final stage derives a temporal graph for each recurring character, with ordered checkpoints, state and development ledgers, epistemic access, and persona evidence (Figure 2f). States cover goals, knowledge, relationships, identity, and constraints. Developments connect a prior state to its catalyst, resulting state, and possible delayed consequence; epistemic links distinguish what the character witnessed, was told, inferred, or does not know. Only information available to a character by a checkpoint may enter its state or knowledge.

Episodes and Storylines provide evidence-linked paths for Task II, while the temporal graph supplies the checkpoint-valid states and knowledge boundaries used by Tasks I and III. These structures remain construction- and evaluator-side. Appendices B.2 and B.5 and Table A8 document the complete schemas, review, validation, and task mapping.

In a five-screenplay pilot, backbone guidance increased mean evidence span from 9.2% to 46.5% of a screenplay and audited reasoning paths from 1.5 to 3.6 logical hops. It also raised cross-scene evidence use from 26.0% to 84.0% and grounded accuracy from 62.5% to 94.2%, while reducing narrative-logic hallucinations from 36.0% to 5.8% (Appendix Table A11). These results establish the backbone's construction benefit within the pilot scope.

3.1.3 Paired work-identity controls

Widely known films may appear in model pretraining data, allowing models to use work-specific identity cues. STAGE pairs each original instance with a title-blind pseudonymized counterpart derived from the frozen entity registry. Reviewed one-to-one mappings propagate through screenplays, task inputs, references, and evaluator evidence while preserving scene order, checkpoints, states, evidence, and answer semantics. Paired differences measure sensitivity to explicit identity cues. Distinctive narrative content may still reveal the underlying work (Yan et al., 2022; Yao et al., 2024). Appendix B.4 specifies the mapping, propagation, and quality-control procedure.

3.2 Tasks

STAGE operationalizes evolving-story understanding as three complementary capabilities: tracking character development, reasoning about narrative evolution, and acting from a checkpoint-valid character state. Internal graph, state, and provenance identifiers remain evaluator-side throughout all tasks. The release contains 434 roles, 2,925 Task I checkpoints and 7,912 states, 5,010 Task II questions, and 5,425 single-turn plus 866 multi-turn Task III instances; Table 1 gives the language breakdown.

The three tasks share evidence, checkpoints, and temporal boundaries by construction. Task I samples successive slices of the character temporal graph and asks models to update the state that holds at each checkpoint. Task II follows evidence-linked Episode and Storyline paths to ask what

Table 1: Overall statistics of STAGE evaluation assets.

Language	Roles	Task I		Task II	Task III	
		Check-points	States	QA	Single-turn	Multi-turn
English	330	2,235	6,070	3,583	4,288	628
Chinese	104	690	1,842	1,427	1,137	238
Total	**434**	**2,925**	**7,912**	**5,010**	**5,425**	**866**

changed, when it became effective, and which consequences persisted across scenes. Task III converts a checkpoint slice into model-visible persona, knowledge, relationship, and memory context, then tests whether those constraints are used in generation. This common provenance supports controlled comparison of state-maintenance, evidence-integration, and grounded-generation failures across tasks.

STAGE is an evaluation-only benchmark. Tasks I–III are released as fixed evaluation sets without training, development, or test splits. The development/test partition in Appendix B.3 belongs exclusively to the separate 20-film validation of construction-time entity resolution.

3.2.1 Task I: Character Development Tracking

Given a focal character's preceding state and observations from the interval between adjacent checkpoints, Task I predicts the current state and newly effective developments across goals, knowledge, relationships, identity, and constraints. **Reference-State Update (RSU)** supplies the reviewed prior state to isolate local updating; **Autoregressive State Update (ASU)** supplies the model's prior prediction to expose accumulated error. Inputs, targets, and prompts otherwise remain fixed, with change, no-change, inaccessible-event, delayed-consequence, and final controls. Full eligibility, construction, and a worked trajectory appear in Appendices B.4 and B.4.

3.2.2 Task II: Cross-Scene Narrative Evolution Reasoning

Task II contains 5,010 questions requiring source-grounded reasoning across scenes. Its six types cover Scene Grounding, Character Understanding, Causal-Motivational Reasoning, Temporal Reasoning, Narrative Progression, and Role-Relation Continuity. Each item has a reference answer and traceable support for a required cross-scene path, following the anti-shortcut motivation of multi-hop QA (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022) (Appendices B.4 and B.4).

3.2.3 Task III: In-Script Character Role-Playing

Task III anchors a focal character and audience request at a screenplay checkpoint. A frozen context pack supplies visible persona evidence, dialogue style, acquired memories, relationships, and the current user turn. Responses must remain natural, in character, and bounded by checkpoint-available knowledge. Single-turn interaction is primary; paired checkpoints and an auxiliary multi-turn protocol test appropriate change, stability, knowledge acquisition, and drift. Construction and a paired example appear in Appendices B.4 and B.4.

3.3 Evaluation

STAGE separates local semantic judgments from deterministic metric aggregation. The judge is independent of the evaluated model and calibrated against blinded human annotations. Confidence intervals resample movies as dependency clusters.

Task I. We report *Current-State Coverage*, *Development Coverage*, *State Contradiction Rate*, and *Development Contradiction Rate*. Current-State and Development Coverage measure how completely a prediction recovers, respectively, the state valid at the checkpoint and the developments that became effective since the preceding checkpoint; the two contradiction rates measure the share of predicted claims in each pool that conflict with screenplay evidence or the checkpoint-valid state. Reference claims receive full, partial, missing, or contradictory labels; full and partial coverage score 1 and 0.5. ASU–RSU Development Coverage measures recursive degradation, and headline scores use movie-macro aggregation (Appendix C.4.1).

Task II. Each generated answer receives a binary correctness judgment against the reference and screenplay evidence. We report Average Accuracy over five generations and Pass@5; citation support remains a separate diagnostic (Appendix C.4.2).

Task III. Each response receives four independent 1–5 scores: Character Fidelity, Memory Faithfulness, Boundary Compliance, and Response Naturalness. Typed checkpoint pairs test expected change, stability, knowledge acquisition, and relationship change; we report the dimensions separately (Appendix C.4.3).

4 Experimental Setup

4.1 Research Questions

RQ1: How accurately can models track character development over time? We test integration of new evidence into current states and developments; RSU–ASU differences isolate error accumulated through recursive state reuse. Separate coverage and contradiction metrics distinguish omission from explicit state conflict.

RQ2: How reliably can models reason about narrative evolution across scenes? We compare flat, hierarchical, and graph-based access on questions about cross-scene change and persistence. Pass@5 measures whether a system can produce at least one correct answer, while average accuracy measures stability across repeated generations.

RQ3: Can models act consistently from a checkpoint-bounded character state? We vary memory access and compare typed checkpoints to test natural, in-character behavior under checkpoint-available information. Independent dimensions distinguish persona fidelity, screenplay-memory use, epistemic boundary compliance, and response quality.

4.2 Models and Baselines

We evaluate Qwen3 (8B, 30B-A3B, and 235B), Llama-3.1 (8B and 70B), GPT-5.5, Gemini 3.1 Pro, and Claude Sonnet 4.6 with standardized prompts (Yang et al., 2025a; Grattafiori et al., 2024; OpenAI, 2026; Google, 2026; Anthropic, 2026). Parameter counts are reported where disclosed by the provider or encoded in the model name; undisclosed counts for proprietary models remain unreported. The open-weight Qwen3 and Llama-3.1 checkpoints are served locally on the single-node cluster described in Appendix B.1, using FP8 or AWQ quantization where required by GPU-memory constraints.

Task I compares RSU and ASU under matched intervals and observations. Task II compares **Hybrid RAG**, which combines dense and lexical scene retrieval, hierarchical **PageIndex** (Zhang et al., 2025) and **A-RAG** (Du et al., 2026), and graph-based access through **GraphRAG** (Edge et al., 2024) and **LightRAG** (Guo et al., 2025b). The Kimi 2.6 full-screenplay reference (Moonshot AI, 2026) uses a different access setting and therefore appears in Appendix C.6.2. All retrieval systems operate over the same screenplay release and reader models, isolating the effects of memory organization and evidence selection.

Task III fixes persona context and varies episodic memory: Persona Card Only, all checkpoint-visible memories, or the top five memories from BM25 or vector retrieval.

4.3 Evaluation Protocol

DeepSeek-V4-Pro (DeepSeek-AI, 2026) serves as the judge for all tasks, independently of the evaluated model; deterministic code converts its judgments into the metrics in Section 3.3. Judge calls use the official DeepSeek API, separately from the local FP8 construction deployment described in Appendix B.1. On 1,000 stratified outputs, it agreed with adjudicated human consensus in 90.3% of cases (Appendix C.3); the two blinded human raters agreed in 91.7% before adjudication. The calibration sample contains 200 Task I state outputs, 500 Task II answers, and 300 open-ended Task III responses, reflecting their different judgment structures. Judge–consensus agreement is 90.2%, 93.0%, and 85.8%, respectively, with Cohen's κ of 0.82, 0.86, and 0.76. The lower Task III agreement reflects its graded, open-ended dimensions.

All systems use the frozen release with temperature $T = 0.2$, top-$p = 0.9$, and at most 8,192 generated tokens. Task I and Task III use one generation per instance; Task II uses five. Prompts, model IDs, decoding settings, hashes, failures, and denominators are logged, and confidence intervals resample movies as dependency clusters.

The model-facing templates, decoding parameters, output budgets, and task-specific memory conditions are fixed uniformly across evaluated models. Prompts exclude benchmark items paired with their gold labels or reference answers as in-context demonstrations. Worked examples in the appendix serve as reader illustrations and are excluded from evaluation inputs.

5 Results and Analysis

Results address tracking, reasoning, and generation on the original benchmark view; paired pseudonymized runs audit sensitivity to explicit work identity.

5.1 RQ1: Character Development Tracking

GPT-5.5 leads both coverage measures under RSU and ASU, while **Claude Sonnet 4.6** has the lowest State Contradiction (Table 2). GPT-5.5 reaches

Model	Reference-State Update				Autoregressive State Update			
	State Cov.	Dev. Cov.	State Contr.	Dev. Contr.	State Cov.	Dev. Cov.	State Contr.	Dev. Contr.
Qwen3-8B	48.20	42.50	2.60	15.10	36.50	31.00	3.80	19.40
Qwen3-30B-A3B	61.80	65.40	1.25	8.10	56.20	60.10	1.40	8.85
Qwen3-235B	66.46	72.11	0.97	6.56	63.39	70.38	0.67	6.66
Llama-3.1-8B	42.10	36.80	3.10	18.50	29.40	23.20	4.90	23.80
Llama-3.1-70B	54.60	52.30	1.85	11.20	47.10	44.80	2.30	13.50
GPT-5.5	**74.50**	**78.80**	0.42	**4.10**	**72.10**	**76.50**	0.45	**4.35**
Gemini 3.1 Pro	60.10	62.80	1.10	7.90	55.30	57.60	1.35	8.40
Claude Sonnet 4.6	68.20	73.50	**0.35**	4.80	65.40	71.00	**0.38**	5.10

Table 2: Task I results under Reference-State Update (RSU) and Autoregressive State Update (ASU) on the original, non-pseudonymized release. All values are movie-macro percentages. Bold marks the best value within each update setting; Contr. denotes Contradiction.

74.50/78.80 Current-State/ Development Coverage under RSU and 72.10/76.50 under ASU; even these maxima leave substantial content uncovered, and low contradiction does not imply complete tracking. Coverage declines from RSU to ASU for every model. With scale, the Development Coverage loss narrows from 11.50 to 1.73 points for Qwen and from 13.60 to 7.50 for Llama, while smaller models produce more contradictions. The protocol gap therefore reflects lost ASU coverage more than additional incompatible claims. Recursive failure appears primarily as silent forgetting: scaling mitigates, but does not remove, the difficulty of maintaining a trajectory across updates.

Pseudonymization shifts RSU and ASU coverage in opposite directions, but no coverage or contradiction change is reliable (all 95% intervals include zero; Appendix Table A18). Memorized identity effects remain unresolved, and between-model differences are descriptive because Table 2 reports point estimates.

5.2 RQ2: Cross-Scene Narrative Evolution Reasoning

Hybrid RAG performs best overall, but the strongest Pass@5 and average accuracy reach only 65.1 and 62.5 (Table 3). The oracle union reaches 78.6%, and Pass@5–accuracy gaps indicate inconsistent evidence reuse. The oracle gain shows that retrieval systems recover complementary evidence, yet no reader model converts it into consistently correct answers across generations. Among alternative structures, **LightRAG** (Guo et al., 2025b) outperforms **GraphRAG** for every model, while **A-RAG** (Du et al., 2026) generally exceeds PageIndex on larger models.

Removing screenplay context lowers accuracy by 41.7–51.6 points, whereas pseudonymization has small, uncertain effects (Appendix Tables A19 and A20). Evidence access thus matters more than explicit identity cues, although semantic content may still reveal the work. Complete gold-scene coverage improves accuracy by 24.0–27.0 points, yet 27.1–29.6% of answers remain incorrect and Temporal Reasoning reaches only 44.9–48.9% (Appendix Table A21). Retrieval coverage is therefore necessary but insufficient for ordered-state reasoning: models still fail to integrate available scenes into the required temporal and causal sequence. The category breakdown also shows retrieval specialization: on Qwen3-235B, LightRAG leads Character Understanding, Role-Relation Continuity, and Temporal Reasoning, whereas Hybrid RAG leads Scene Grounding, Causal-Motivational Reasoning, and Narrative Progression (Appendix Table A31). No structure dominates all reasoning types; representative temporal and causal failures appear in Table A29.

5.3 RQ3: Grounded Persona Consistency in In-Script Character Role-Playing

Task III separates surface persona imitation from screenplay-grounded role-playing. Persona-only inputs remain fluent and in character, yet Memory Faithfulness is only 1.53–1.82; full memory raises it to 3.62–4.96 for every model (Table 4). BM25 instead leads Fidelity, Naturalness, and Boundary Compliance, with vector retrieval close behind. Fluent characterization and accurate use of screenplay memory are therefore distinct capabilities. BM25 concentrates generation on a small set of locally relevant memories, supporting style and situation fidelity, but its top-five budget can omit facts required for strict grounding; full memory

Model	Hybrid RAG		PageIndex		A-RAG		GraphRAG		LightRAG	
	Pass@5	Avg. Acc.	Pass@5	Avg. Acc.	Pass@5	Avg. Acc.	Pass@5	Avg. Acc.	Pass@5	Avg. Acc.
Qwen3-8B	46.1	42.4	34.9	18.0	27.7	23.6	31.8	16.7	42.9	36.6
Qwen3-30B-A3B	55.3	51.5	34.2	33.7	37.2	30.2	41.2	29.8	54.2	38.8
Qwen3-235B	63.7	60.8	38.6	34.3	48.6	36.0	47.1	33.5	60.9	43.9
Llama-3.1-8B	41.3	38.1	25.2	16.8	25.7	21.8	29.6	19.8	41.5	33.9
Llama-3.1-70B	55.8	51.6	35.8	30.2	42.4	33.9	43.3	30.1	52.0	38.7
GPT-5.5	65.0	**62.5**	40.6	35.6	**53.7**	**42.1**	**51.7**	39.2	**61.6**	**47.9**
Gemini 3.1 Pro	63.9	61.2	40.6	35.6	51.4	40.6	51.5	**40.6**	60.2	46.5
Claude Sonnet 4.6	**65.1**	61.6	**44.1**	**40.0**	52.4	41.2	50.3	39.3	60.1	46.4

Table 3: Cross-Scene Narrative Evolution Reasoning results on the full, original Task II question-answering release. Pass@5 counts a question as correct if any of five generations is judged correct; Average Accuracy averages correctness across the five generations.

Metric	Qwen3-8B				Qwen3-30B-A3B				Qwen3-235B				Llama-3.1-8B			
	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect
Fidelity	4.33	4.51	**4.69**	4.68	4.42	4.62	**4.78**	4.77	4.52	4.72	**4.83**	4.83	3.98	4.12	**4.35**	4.35
Natural.	4.72	4.79	**4.82**	4.82	4.82	4.89	**4.92**	4.92	4.84	4.93	**4.95**	4.95	4.65	4.72	**4.77**	4.76
Boundary	3.86	4.25	**4.38**	4.38	4.26	4.55	**4.68**	4.68	4.63	4.79	**4.89**	4.88	3.69	4.08	**4.21**	4.21
Mem. Faith	1.58	**3.85**	3.45	3.37	1.68	**4.32**	3.87	3.78	1.76	**4.69**	4.19	4.10	1.53	**3.62**	3.24	3.17
Metric	**Llama-3.1-70B**				**GPT-5.5**				**Gemini 3.1 Pro**				**Claude Sonnet 4.6**			
	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect
Fidelity	4.32	4.68	**4.85**	4.85	4.76	4.88	**4.97**	4.96	4.69	4.81	**4.93**	4.92	4.71	4.83	**4.93**	4.92
Natural.	4.75	4.83	**4.87**	4.86	4.91	4.98	**4.99**	4.99	4.88	4.95	**4.97**	4.96	4.90	4.96	**4.99**	4.98
Boundary	4.13	4.44	**4.57**	4.57	4.82	4.93	**4.98**	4.98	4.69	4.82	**4.91**	4.90	4.76	4.88	**4.95**	4.95
Mem. Faith	1.65	**4.18**	3.74	3.66	1.82	**4.96**	4.44	4.34	1.79	**4.81**	4.31	4.21	1.80	**4.88**	4.37	4.27

Table 4: Complete Task III single-turn results across all eight models and four memory settings on the original release (1–5 scale). Bold marks the best setting per model and metric. PC: Persona Card Only; FM: Full Memory Context; Vect: vector retrieval.

supplies those facts while introducing more competing episodes. Selective retrieval therefore favors persona expression, whereas complete checkpoint-valid context favors narrative recall. The consistency of this trade-off across models means that no single access strategy dominates all four objectives; separate dimension-wise scores prevent either behavior from being hidden by an aggregate.

The multi-turn test retains the full-memory advantage but shows model-specific drift (Appendix Table A32). Support diagnostics further distinguish retrieving a relevant memory from using it correctly (Appendix C.6.4), confirming that access alone does not ensure grounding. Identity audits leave Memory Faithfulness, Naturalness, and multi-turn continuity broadly stable, with small Fidelity and Boundary Compliance shifts (Appendix Tables A22 and A23). These effects remain diagnostic rather than a general contamination estimate. Representative failures expose the remaining gap: dialogue can sound natural while inventing unsupported premises or ignoring checkpoint-valid interaction context (Appendix Table A30).

6 Conclusion

STAGE evaluates full-screenplay understanding through three aligned capabilities: tracking character development, reasoning across scenes, and role-playing within checkpoint-valid state and knowledge boundaries. Across 151 English and Chinese films, the tasks expose complementary failures in maintaining, integrating, and applying evolving narrative state. Recursive updates lose prior developments, cross-scene QA remains difficult even with complete annotated evidence, and persona expression can diverge from screenplay faithfulness. These findings show that context length, retrieval coverage, and fluent generation are incomplete proxies for narrative understanding. Together, the shared provenance-linked backbone supports diagnosis of state updating, evidence integration, memory, epistemic control, and identity effects.

7 Limitations

STAGE has three main limitations.

Construction quality and cost. The backbone and task assets use a hybrid human–LLM pipeline, so residual annotation errors may remain, particularly in Tasks I and III where audits are representative. Task II receives exhaustive review, while risk-based checks, source-level provenance, and deterministic validation make all retained items traceable. These controls prioritize release quality, although construction cost still limits throughput.

Context formulation. Performance depends partly on how evidence is organized and accessed. Task I uses fixed checkpoint intervals, Task II compares retrieval systems, and Task III varies checkpoint-visible memory. STAGE standardizes prompts and access conditions within each comparison and reports retrieval and memory ablations. The resulting claims therefore concern model behavior under explicit, documented context conditions.

Coverage and identity. The corpus covers English and Chinese movie screenplays, with fewer Chinese films because available sources are less centralized and less consistently formatted. The bilingual release broadens language coverage, while the source index and reconstruction tools provide a basis for expansion. The released pseudonymized view reduces explicit title and entity cues, although distinctive semantic content may still reveal a work. Paired differences thus quantify sensitivity to work identity; estimating training-data contamination requires separate evidence.

8 Ethical Considerations

STAGE follows the linked-source release design of NarrativeQA (Kočiský et al., 2018). The public package contains source metadata and links, integrity information, an original-identity reconstruction tool, STAGE-authored annotations, and evaluation code. Users obtain screenplay text from its recorded source and materialize the original-identity view locally. The public package also includes the pseudonymized control used for the work-identity robustness results. The STAGE-authored assets are provided for non-commercial research with source attribution and a contact and removal procedure; ownership and commercial rights remain with the source-work holders.

Preprocessing removes source front matter and incidental contact information, and derived assets are screened before release. All annotation was conducted voluntarily by adult coauthor domain experts. Because screenplays may contain mature fictional content, the dataset card flags this possibility and recommends appropriate access controls for research use.

LLMs assist structured annotation and evaluation, with source provenance, human review, deterministic validation, and judge calibration providing quality controls. STAGE is intended for controlled research evaluation of narrative understanding. Its task-specific metrics are diagnostic signals, with broader literary judgment beyond their scope.

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A Dataset Details

A.1 Data Collection, Annotation, and Release Policy

STAGE is built from full-length movie screenplays collected from publicly available sources. English scripts are primarily gathered from online screenplay repositories indexed by IMDb (IMDb, 2024), while Chinese scripts are collected from publicly accessible PDF or Word documents. These Chinese sources are often substantially less standardized in structure, with variation in scene markers, title versus numeric indexing, and the presentation of dialogue and action lines. We exclude plot summaries, transcripts, and fan-written adaptations, and retain only screenplay texts that preserve scene boundaries, dialogue, and action descriptions.

Release package and governance. The public release contains a source index, retrieval and integrity metadata, the local reconstruction tool, STAGE-authored annotations, and evaluation code. Users obtain screenplay text from its recorded source, after which the tool creates the scene-structured original-identity view while preserving names and title cues. The work-identity analysis reported in this paper uses the released pseudonymized counterparts; private identity mappings remain author-side.

Each source entry records available provenance, retrieval metadata, and integrity information. Each locally processed screenplay also has validation records. The package distinguishes source metadata, STAGE-authored scene normalization and structural annotations, derived narrative and task assets, and evaluation code. The STAGE-authored assets are released for non-commercial research with source attribution and a documented contact and removal procedure.

Source preprocessing removes front matter and incidental contact metadata, including cover pages, headers, contact details, agency addresses, and signatures, before downstream extraction. Derived entity records and QA assets are additionally screened for residual contact information before release.

Genre	Count	Prop.	Genre	Count	Prop.
Drama	78	17.0	Horror	13	2.8
Comedy	59	12.9	Family	12	2.6
Crime	42	9.2	Sports	11	2.4
Action	38	8.3	War	8	1.7
Thriller	38	8.3	Social	7	1.5
Romance	33	7.2	Mystery	6	1.3
Adventure	23	5.0	Social Issues	6	1.3
Fantasy	21	4.6	Music	5	1.1
Science Fiction	19	4.1	Youth	5	1.1
History	15	3.3	Others	20	4.4

Table A1: Genre distribution in STAGE after taxonomy consolidation. Statistics are aggregated across all genre annotation slots (up to three per movie) over both English and Chinese films.

Seven bilingual coauthor-annotators provide human oversight, drawing on professional expertise in screenwriting, directing, and narrative consulting. LLM extraction and generation use fixed schemas and prompts; annotators exhaustively review provisional Task II items and audit representative Task I claims and Task III interactions, with additional review of low-confidence outputs, schema violations, and logical inconsistencies. The same criteria are applied to English and Chinese screenplays; detailed audit coverage appears in Appendix B.5.

A.2 Corpus and Backbone Statistics

Movie Genre Distribution. Each movie is annotated with up to three genre labels used only as descriptive metadata. To reduce sparsity, low-frequency stylistic variants are merged into parent genres. Table A1 reports the aggregated distribution across all genre annotation slots, so counts do not sum to the number of movies.

Release-Year Distribution. The 151 screenplays span releases from 1931 to 2023. The English subset ($n = 109$) ranges from 1931 to 2015, with a median release year of 2001; the Chinese subset ($n = 42$) ranges from 1979 to 2023, with a median of 2011.5. Figure A1 reports decade-level proportions within each language, rather than raw counts, because the two subsets differ in size. These metadata characterize the corpus's temporal coverage; the paired pseudonymization and closed-book analyses provide the direct tests of work- identity effects.

Screenplay Length and Scene Count Distribution. We measure screenplay length as the total word count of scene titles, subtitles, and contents in the locally materialized screenplay records,

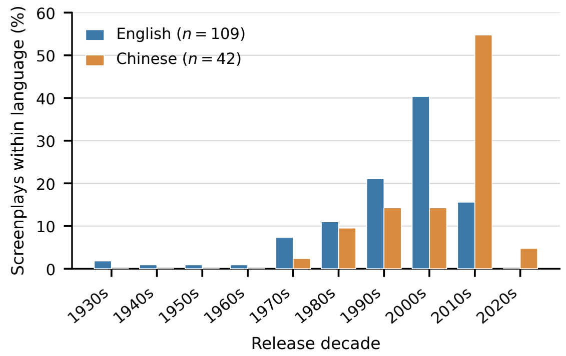

Figure A1: Release-decade distribution of STAGE screenplays. Bars show the percentage within each language subset to account for their different sizes. All 151 screenplays have valid release-year metadata.

and structural complexity as the number of scene records. Table A2 reports summary statistics by language; screenplay lengths range from 12,702 to 70,832 words and scene counts from 12 to 600.

Backbone Schema. The canonical KG uses a fixed seven-type entity registry (Table A3). Events and Interactions remain evidence-bearing records outside the KG node registry and are later assigned to Episodes. Occasions serve as reusable narrative contexts and may therefore be canonical entities and relation endpoints. KG relations link canonical entities under a fixed predicate registry and retain the validation fields summarized in Table A4.

Graph Size Statistics. Table A5 summarizes the canonical KG, induced narrative units, Episodes, and Evolution Storylines per screenplay in the full release.

A.3 Task Release Statistics

Character Development Tracking Assets. Task I release statistics distinguish movies, focal-character trajectories, and formal checkpoint instances. Movies define source documents and dependency clusters; focal-character trajectories are the primary aggregation units; and formal checkpoints are atomic scored observations. For each focal character, Task I contains an ordered checkpoint sequence, fixed adjacent-checkpoint intervals, and evaluator-side rubrics grounded in state, development, invariant, and future-negative evidence. Runtime processing produces the interval memory exclusively for evaluation. Checkpoint controls include actual changes, no-change intervals, inaccessible events, delayed consequences, and final states. Their types remain explicit in the released checkpoint metadata, supporting stratified analysis separately from corpus, trajectory, and context-management counts.

Cross-Scene Evolution Question Taxonomy. Task II uses the six-category narrative-reasoning taxonomy in Table A6. The categories and counts match the released 5,010-question package used in our experiments.

Reference answers are concise. English answers average 9.51 words (median 8; range 1–50), while Chinese answers average 18.11 characters (median 15; range 2–116). We report language-appropriate units because whitespace-based word counts are not comparable for Chinese text.

In-Script Character Role-Playing Assets. Task III contains checkpoint-bounded single-turn interactions and paired cases for evolving, invariant, inaccessible, and post-disclosure conditions. Legacy multi-turn interactions are retained as an auxiliary stress test; the released counts are reported in Table 1.

B Benchmark Construction Details

This appendix details screenplay preprocessing, narrative-backbone construction and validation, and task-specific asset generation.

B.1 Data Preprocessing and Computational Environment

Because screenplay formatting is heterogeneous, preprocessing combines rule-based parsing, LLM assistance, and targeted human verification. English scripts use regex-based scene-boundary detection with LLM correction for ambiguous cases, while Chinese scripts often require OCR followed by heuristic and LLM-assisted scene recovery. For both languages, the final output is a sequence of stable scene identifiers used for downstream grounding.

Token-Budgeted Scene Segmentation. The scene is the primary ordering and provenance unit throughout STAGE. Complete consecutive scenes are packed under the measured tokenizer budget whenever possible. A scene is partitioned only when its token sequence alone exceeds the budget; in that case, tokenizer offsets define ordered, non-overlapping scene-parts. No character-based truncation, semantic summarization, or silent text

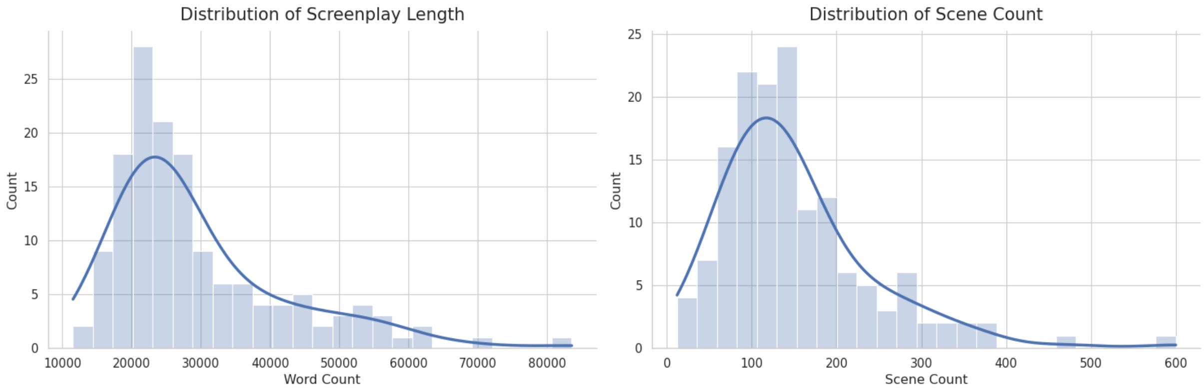

Figure A2: Distribution of screenplay lengths and scene counts in STAGE. The left histogram shows the distribution of screenplay word counts, while the right histogram shows the distribution of scene counts across all movies in the dataset.

		Screenplay Length (words)			Scene Count		
Language	**#Movies**	**Min**	**Mean**	**Max**	**Min**	**Mean**	**Max**
English	109	14,138	29,506	51,937	18	167	600
Chinese	42	12,702	31,530	70,832	12	110	373

Table A2: Summary statistics of screenplay length (in words) and scene count for English and Chinese films in STAGE.

Entity Type	Description
Character	Concrete roles in the story, including human or anthropomorphized characters.
Occasion	Concrete activities or situations that can organize multiple events, such as meetings, confrontations, investigations, or ceremonies.
Location	Physical, geographical, or virtual places and scenes where events occur.
TimePoint	Specific points in time or time spans referenced in the narrative.
Object	Key items, props, or devices that play an important role in the story.
Concept	Abstract categories, ideas, species, or other non-organizational concepts.
Organization	Institutions, factions, companies, agencies, or other organized groups that can participate in narrative relations.

Table A3: Entity types in the shared narrative backbone schema.

deletion is permitted. Before a scene or block enters extraction or a model-facing Task I call, the runner verifies that concatenating its parts reconstructs the original token sequence exactly. Scene and part indices are retained so every derived record can be mapped back to the original text.

Runtime environment. The model-assisted construction pipeline uses **DeepSeek-V4-Pro** (DeepSeek-AI, 2026) for narrative-unit and KG extraction, Episode/Storyline induction, character-temporal asset construction, and Task I–III generation, using identical criteria for English and Chinese. DeepSeek-V4-Pro is deployed locally in FP8 precision through vLLM with bounded concurrency. All local preprocessing, dataset construction, bge-m3 embedding, deterministic validation, and open-weight model inference are performed on a single node with eight NVIDIA A100-SXM4 GPUs (80 GB each), driver 535.129.03, and CUDA 12.4. Evaluated Qwen3 and Llama-3.1 checkpoints run on the same node through vLLM, using FP8 or AWQ quantization as needed to fit within GPU memory. Qwen3 variants serve exclusively as evaluated systems. The released reference environment targets Python 3.12 and records all direct package dependencies and minimum versions.

B.2 The Narrative Backbone Construction Pipeline

Extraction Paradigm Choice. STAGE uses schema-driven KG construction (Zhang and Soh, 2024) with a fixed ontology, keeping narrative units, entity types, and relation predicates comparable across screenplays. Incompatible candidates are excluded or reviewed against this ontology.

Field	Constraint
Canonical endpoints	Both endpoints must resolve to canonical entity identifiers; unresolved surface names and Event/Interaction records cannot serve as KG endpoints.
Predicate	The predicate must belong to the frozen registry and satisfy its domain and range constraints. The taxonomy is not expanded in response to extraction errors.
Qualifiers	Polarity, modality, and temporal scope are stored explicitly when supported by the screenplay.
Source evidence	Every retained relation stores source-scene identifiers and original-text evidence sufficient for later verification.
Deduplication key	Relations with identical canonical endpoints, predicate, qualifiers, and temporal scope are merged deterministically while preserving all evidence IDs.

Table A4: Validation schema for canonical KG relations. The frozen predicate inventory and domain–range registry are released with the construction configuration.

	English ($N = 109$)		**Chinese** ($N = 42$)	
Asset / Metric	**Mean**	**Min–Max**	**Mean**	**Min–Max**
Canonical KG Assets				
Canonical Entities	346.1	222–431	246.3	142–304
Semantic Relations	145.8	69–201	71.0	48–95
Narrative Graph Assets				
Atomic Narrative Units†	483.0	285–578	330.0	184–366
Episodes	175.3	81–230	134.0	74–196
Evolution Storylines	10.4	4–15	7.7	3–12

Table A5: Summary statistics of canonical KG and narrative-graph assets per screenplay across the full STAGE benchmark, reported after entity canonicalization and graph validation. †Atomic narrative units aggregate evidence-bearing Events, Interactions, and Occasions.

Why Staged Extraction. Extraction proceeds from narrative units to seven-type entities and relations over identified entities, followed by scene-level review and global canonicalization. Each stage must satisfy source-grounding and schema constraints before entering the next. Appendix B.3 compares this complete staged pipeline with a one-pass extraction baseline.

Narrative units and KG boundary. Events, Interactions, and Occasions retain their participants, narrative attributes, and original-text spans as source-grounded nodes in the final narrative graph. Semantic KG relations use canonical entity endpoints. Event and Interaction records remain evidence-bearing narrative nodes, while reusable Occasions also appear in the entity registry.

Entity normalization and canonicalization. Surface forms are normalized before type-aware lexical and semantic alias matching, with multiple text-supported types allowed (e.g., Organization–Location when an institution also functions as a place). Resolved references and Episodes inherit canonical entities; the complete procedure appears below.

Algorithm 1 Token-Budgeted Processing of Ordered Scenes

Require: Ordered scenes $\mathcal{S}$, tokenizer T, input budget τ
Ensure: Ordered blocks $\mathcal{B}$ with complete source coverage

```
Initialize B ← [ ] and current block b ← ∅
for all s_i ∈ S do
    Tokenize the full scene: u_i ← T(s_i)
    if |u_i| > τ then
        Flush non-empty b to B and set b ← ∅
        Split u_i at tokenizer offsets into ordered parts p_{i,1}, ..., p_{i,m}
        Verify p_{i,1} || ··· || p_{i,m} = u_i
        Append the scene-parts to B with scene and part indices
    else if |T(b || s_i)| ≤ τ then
        Set b ← b || s_i
    else
        Append b to B and set b ← s_i
    end if
end for
if b ≠ ∅ then
    Append b to B
end if
Verify ordered, non-duplicated coverage of every source scene
return B
```

Relation consolidation and quality control. Candidates require resolved endpoints, a frozen predicate satisfying its domain–range constraints, and source evidence. Retained records preserve

Question Type	Count	Primary Reasoning Demand
Scene Grounding	1,207	Locate the relevant scene or scenes and identify the action, event, setting, or object grounded there.
Character Understanding	1,103	Infer a character's local or evolving goals, intentions, knowledge, states, and decisions from screenplay evidence.
Causal-Motivational Reasoning	1,041	Connect an action or development to its supported cause, trigger, or character motivation.
Temporal Reasoning	766	Resolve event order, beginnings and endings, before–after relations, or the point at which a development occurs.
Narrative Progression	597	Identify how turning points and earlier developments initiate, redirect, or shape a later plot trajectory.
Role-Relation Continuity	296	Track social roles and relationships across scenes, including their persistence, escalation, weakening, or transformation.

Table A6: Narrative-reasoning taxonomy and category counts for the released STAGE Task II package (5,010 questions).

Category	Resource and execution scope	Compute / requests	Tokens (input/output)	Est. cost
Data construction	Local 8×A100 (80 GB); backbone and task-asset extraction for 151 screenplays	≈ 240 GPU-h	≈ 250M in ≈ 20M out	Internal compute
Open-weight evaluation	Local 8×A100 (80 GB); Tasks I–III with Qwen3 and Llama-3.1	≈ 320 GPU-h	≈ 450M in ≈ 40M out	Internal compute
Hosted model evaluation	Official OpenAI, Anthropic, Google, and Moonshot APIs; GPT-5.5, Claude Sonnet 4.6, and Gemini 3.1 Pro on Tasks I–III; Kimi 2.6 on the supplementary Task II full-context baseline	≈ 36,000/model	≈ 150M in ≈ 15M out per model	$450–600 per model
Automated judge	Official DeepSeek API; semantic scoring for Tasks I–III	≈ 120,000 total	≈ 90M in ≈ 10M out total	≈ $20 total

Table A7: Estimated computational budget and API usage. Local construction and open-weight evaluation used approximately 560 A100 GPU-hours in total. Hosted model evaluation cost approximately $2,200, and automated judging cost approximately $20. Across the listed local and hosted workloads, estimated usage totals approximately 1.39B input tokens and 130M output tokens. Token, request, and cost values are rounded estimates.

polarity, modality, temporal scope, and evidence IDs; records with identical endpoints, predicates, qualifiers, and time ranges merge their evidence. Logged retries and targeted repairs handle failures without dropping source coverage; detailed controls appear below.

Episode and Storyline induction. Each Event or Interaction is assigned to exactly one primary, scene-local Episode. An Episode summarizes a setup, development, and outcome, where the outcome may remain open; scene structure determines the number of Episodes. Candidate Episode relations are considered only within a scene-distance window of eight and must share an interpretable bridge, such as a participant, location, Occasion, interaction endpoint, state target, open thread, or aligned trigger and result. Retained forward relations are `causes`, `enables`, `continues`, `resolves`, `reverses`, and `reveals`. Evolution Storylines organize connected cross-scene transitions in character, relationship, goal or conflict, knowledge or belief, and social or institutional development. Every transition records a before state, catalyst, after state, and Episode/relation provenance.

Character temporal graph. We convert the objective narrative hierarchy into a character-relative graph containing evidence units, a character registry, state ledgers, development records, epistemic-access records, persona and style evidence, and checkpoint sequences. States have explicit validity intervals; developments link a before state, catalyst, resulting state, and observable consequence; epistemic access distinguishes information that was witnessed, involved the character, was told, was inferred, or remains unknown. Checkpoints are anchored to a scene and a within-scene character position and include change, no-change, inaccessible, delayed-consequence, and final cases. Generated summaries may propose candidates, but only original screenplay evidence can support a retained record.

Detailed entity resolution procedure. STAGE normalizes chunk-level entity mentions before disambiguation and consolidation.

Entity normalization and scope resolution. Entity mentions are first checked for unresolved scope, type inconsistency, or ill-defined references. An LLM refines names and types when needed; mentions that cannot be normalized reliably are left separate and excluded from merging.

Lexical-overlap filtering and embedding similarity. After normalization, we identify potentially ambiguous entity groups using type-aware lexical-overlap heuristics over names and aliases. This high-recall stage captures surface-form variants before semantic matching. It is especially useful when a character appears under a full name in some chunks and a first or last name in others. Each remaining entity candidate i is then represented by two embeddings: a name embedding $e_i^{(n)}$ and a description embedding $e_i^{(d)}$. Pairwise cosine similarities are computed as $S_{ij}^{(n)} = \cos(e_i^{(n)}, e_j^{(n)})$ and $S_{ij}^{(d)} = \cos(e_i^{(d)}, e_j^{(d)})$, and combined as

$$S = \alpha S^{(n)} + (1 - \alpha) S^{(d)},$$

where α controls the relative contribution of surface-form similarity and descriptive semantics.

k-NN graph construction and cluster estimation. We construct a symmetric k-nearest-neighbor graph under the combined similarity S, with adjacency matrix A and degree matrix $D = \mathrm{diag}(A\mathbf{1})$, and form the graph Laplacian $L = D - A$. Let $\lambda_1 \leq \cdots \leq \lambda_n$ denote the eigenvalues of L. The number of clusters is estimated using an eigengap heuristic over non-trivial modes, with additional regularization in small-sample regimes to avoid over-aggressive merging.

Clustering and merge candidate generation. Each entity is embedded into a joint representation $\tilde{e}_i = [\beta e_i^{(n)} \; ; \; (1 - \beta) e_i^{(d)}]$, and k-means clustering is applied using the estimated cluster count. Clusters containing a single entity are discarded, and only multi-entity clusters are treated as merge candidates. The final candidate pool is the union of lexical-overlap matches and embedding-cluster matches, which biases the system toward recall before adjudication.

LLM-assisted disambiguation and application. Entity resolution uses two semantic stages after deterministic candidate filtering. The first stage classifies retrieved mention pairs as `same_identity`, `different_identity`, or `uncertain`. High-confidence positive decisions must satisfy type, scope, same-scene, and accumulated negative constraints. The second stage reviews newly adjacent cluster pairs using their complete names, types, scopes, timelines, and source evidence. We run at most two cluster-review rounds and stop when a round produces no merge. Uncertain cases remain separate. Accepted merges produce a canonical name and alias set that is applied to entity nodes and relation endpoints throughout the knowledge graph. Decisions use names, aliases, types, scopes, temporal continuity, and source evidence. Surface similarity and generated rationales serve as audit signals; retained decisions require source evidence.

Detailed extraction quality control and post-processing. STAGE applies three layers of quality control during knowledge-graph construction: reflection-based acceptance with bounded retries, deterministic rule-based post-processing, and provenance-preserving human review.

Reflection-based acceptance with bounded retries. Each extraction run is followed by an LLM-based reflection step that assigns a 0–10 quality score and decides whether the result should be accepted or re-extracted. Reflection evaluates:

- **Accuracy**: logical validity of extracted triples, adherence to the fixed entity–relation schema, and absence of malformed structures (e.g., invalid subject–object assignments or undefined relation types);
- **Consistency**: stability of entity naming and typing, including detection of referential ambiguity or near-duplicate entities;
- **Redundancy**: identification of low-value, repetitive, or narratively irrelevant entities and relations.

Critical errors such as schema violations or self-referential relations force the score below threshold.

Construction asset	Task I: tracking	Task II: evolution reasoning	Task III: role-playing
Episode/Storyline hierarchy	Supplies evidence-linked change and turning-point candidates that are resolved into character-relative records.	Supplies multi-scene transition paths, causal or continuity links, and compact support-scene candidates.	Groups character experiences into episodic-memory candidates while preserving their source Episode and scene provenance.
Character temporal graph	Defines ordered checkpoints and evaluator-side current-state and interval- development targets.	Provides before/after contrasts, effective times, epistemic transitions, and no-change or delayed-consequence controls.	Filters persona, memory, and relationship evidence at each checkpoint and defines evaluator-side unknown and future-negative constraints.

Table A8: How the two higher-level backbone assets support task construction. The table describes construction- and evaluator-side dependencies. Evaluated models receive the task-specific inputs defined in Section 3.

We accept only outputs above the fixed threshold and otherwise re-run extraction with reflection feedback up to a bounded retry budget, retaining the highest-scoring result.

Deterministic Rule-Based Post-processing. After extraction, deterministic post-processing corrects systematic schema-level errors, enforces structural validity, and reduces redundancy. We apply three classes of rule-based operations to predicted relations:

- **Endpoint validation.** Each relation is validated against the extracted entity inventory. Relations with missing, unresolved, or dangling subject or object entities are discarded in strict acceptance mode.
- **Schema-constrained type repair.** Each relation is checked against admissible subject–object type constraints defined by the shared narrative backbone schema. If a predicted relation violates the schema but the endpoint entity types uniquely determine a single valid relation type, the relation type is rewritten to that canonical type. If multiple candidate relation types are possible or no valid type exists, the relation is removed under strict mode.
- **Global deduplication.** Relations are globally deduplicated by the tuple (*subject*, *object*, *relation type*). When duplicates occur, the variant with richer or more specific descriptive content is retained.

For example, Character–Location and Character–Organization endpoint pairs map to `located_at` and `affiliated_with`, while event–location and event–time pairs map to the corresponding occurrence relations. Ambiguous or schema-invalid cases are removed rather than repaired heuristically.

On the entity side, STAGE also applies type-aware name deduplication. When an abstract `AtomicEvent` node and a concrete narrative entity share a surface name, the non-event entity is retained.

Provenance-preserving human review and correction. Human review covers both low-confidence KG fragments and movie-level graph structure. After model-assisted construction, reviewers audit entity canonicalization, Episode semantics, inter-Episode relations, Storyline eligibility, temporal task prompts, and QA items against the cited screenplay scenes. Intervention targets errors that would change benchmark semantics: source-format contamination, confusion between a character's claim and narrative fact, causal or intentional interpretations unsupported by observable evidence, transient states promoted to development Storylines, incorrect actor or checkpoint assignments, and task items that are duplicative or not uniquely answerable.

Each accepted decision is encoded as a typed patch against stable artifact IDs. The patch records the action, reason, and supporting scene IDs, while both the superseded model artifact and the reviewed output remain available for audit. Corrections therefore operate on the smallest affected structure—for example, dropping or merging an entity, revising an Episode field, deleting an unsupported Episode relation or Storyline, rejecting an invalid role-playing prompt, or replacing a QA item—rather than overwriting the original artifact or repeatedly querying the construction model until it returns a preferred answer.

Figure A3 makes this process concrete by expanding one Episode-level correction from source evidence through the applied field updates, then summarizing representative patch actions across all six reviewed construction layers.

B.3 Pipeline Validation and Efficiency Analysis

Annotation workflow. The narrative backbone reduces the open-ended search required for annotation. In pilot construction, experts designing cross-scene questions or character trajectories had to track dispersed entities, events, relations, and evidence throughout each screenplay. With the backbone, annotators start from linked entities, extracted events, induced episodes, inter-episode relations, and source provenance, then verify, refine, or reject candidates against the screenplay. This workflow preserves human judgment while making evidence checks more systematic and consistent across Task I and Task II assets.

Held-out entity-resolution validation. We evaluate canonicalization in a held-out 20-film study. Six movies (three English and three Chinese) form the development split; the remaining 14 form a disjoint test split. The development movies are used only to select the embedding-graph threshold (0.86), which is then frozen. Seven domain experts with screenwriting or directing experience annotated the candidate units: two independently assigned entity profiles to canonical clusters, and a third adjudicated disagreements. Sampling emphasizes difficult cross-scene identity cases, including aliases, titles and descriptive references, same-name entities, and fine-grained location or object boundaries.

We compare normalized exact-name matching, an embedding-only candidate graph, and the complete STAGE pipeline. On the common English Character scope, we also evaluate Maverick (Martinelli et al., 2024) in clustering-only mode with predefined mention boundaries. Each profile supplies its surface name, scene identifier, description, and one evidence sentence. The bilingual Full Schema panel excludes Maverick because that setting requires canonicalization across all seven STAGE entity types, beyond its English coreference scope. We report pairwise precision, recall, and F1; B^3, $CEAF_e$, and CoNLL F1; and merge and split errors. Confidence intervals use 10,000 movie-cluster bootstrap replicates.

STAGE performs best in both panels of Table A9, including bilingual canonicalization across all seven entity types. The study isolates clustering under fixed profile boundaries; end-to-end mention detection and release-wide entity recall remain outside its scope.

Pilot Comparison with One-Pass Extraction. Using DeepSeek-V4-Pro on the shared five-movie pilot set, we compare two end-to-end construction procedures. The *One-pass Baseline* prompts the model to extract all seven entity types and fifteen relation types in a single pass. The *STAGE Staged Extraction Pipeline* separates extraction into restricted passes and then applies reflection-based retries, endpoint validation, schema-constrained repair, and provenance-preserving human review. Table A10 reports average extraction outcomes. Because these procedures differ in both decomposition and subsequent quality controls, this pilot evaluates the complete pipelines rather than the isolated effect of staging.

Coverage and schema compliance. The complete STAGE pipeline produced 3.5× as many entity nodes, 5.9× as many relation triples, and 3.64× as many nodes per chunk as the one-pass baseline. Its schema-violation rate was also lower (5.2% versus 23.8%). These results show that the complete pipeline yields denser, more schema-compliant extraction outputs on this pilot; they do not attribute the differences to staged decomposition alone.

Impact of narrative backbone on benchmark quality. We evaluate the downstream benefit of the narrative backbone on five representative screenplays, three English and two Chinese. We compare two construction modes: **Raw-Text Generation**, where items are generated directly from screenplay text, and **Backbone-Guided Generation**, where item construction uses the STAGE narrative backbone with episode links and source-scene provenance.

Table A11 summarizes the audit results. Temporal span is measured as the physical distance between the evidence points required for a correct response, normalized by screenplay length. Logical hops count the number of atomic narrative facts traversed in the manually audited reasoning path. Grounded accuracy and hallucination rate are judged against screenplay evidence, with hallucination denoting plausible but unsupported narrative logic, external lore, or future information unavailable at the queried point.

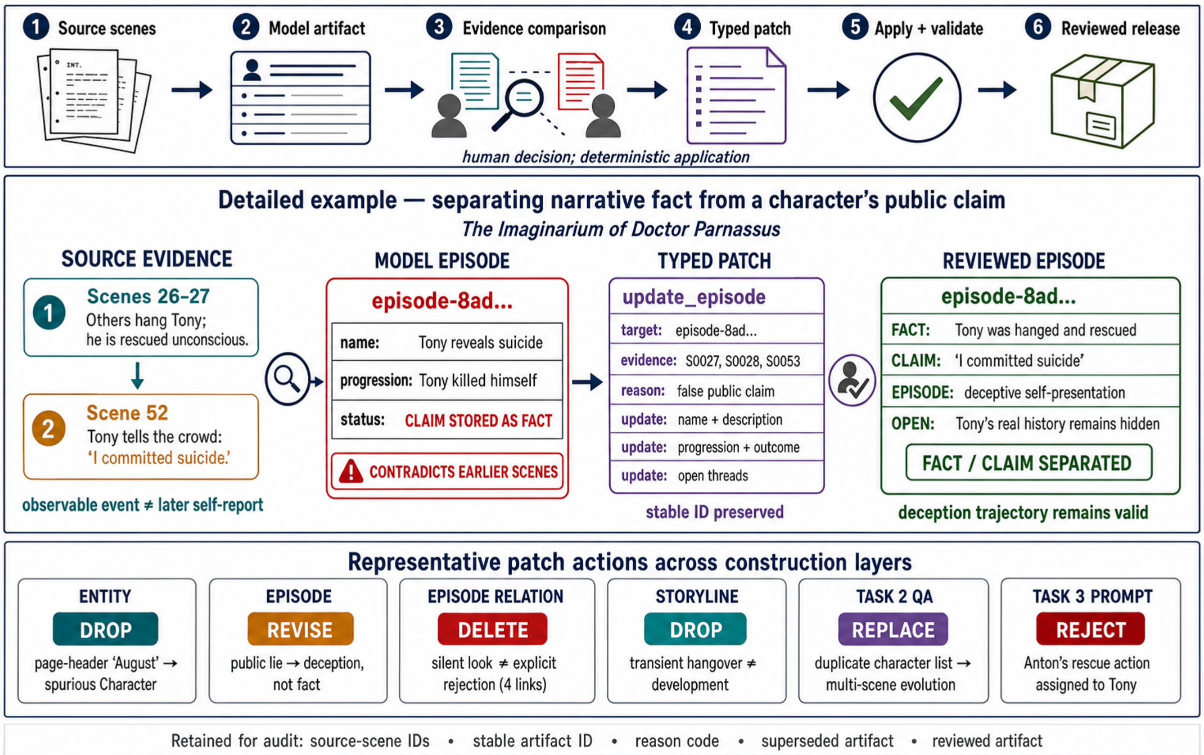

Figure A3: Provenance-preserving human review in STAGE. The example traces an Episode-level correction from cited scenes to a typed, validated patch; the bottom band summarizes representative actions across six reviewed construction layers.

Method	Prec.	Rec.	Pair F1 [CI]	B^3 F1	CEAF$_e$ F1	CoNLL F1 [CI]	Merge	Split
Panel A: English character profiles								
Normalized string	**100.0**	51.2	67.7 [58.3, 74.6]	78.0	60.6	73.8 [66.6, 78.9]	**0**	84
Embedding graph	93.8	17.4	29.4 [2.5, 53.1]	55.0	29.3	40.6 [25.4, 54.6]	2	142
Maverick	**100.0**	87.2	93.2 [86.8, 98.3]	94.7	86.5	92.5 [86.8, 98.4]	**0**	22
STAGE Full	99.4	**100.0**	**99.7 [99.1, 100.0]**	**99.5**	**96.6**	**98.5 [95.8, 100.0]**	1	**0**
Panel B: Bilingual seven-type schema								
Normalized string	94.9	56.0	70.4 [64.9, 75.1]	83.1	68.4	78.2 [73.6, 81.2]	**8**	118
Embedding graph	71.0	16.4	26.7 [11.2, 42.7]	62.3	35.9	44.5 [37.1, 53.2]	18	224
STAGE Full	**95.0**	**98.5**	**96.7 [93.6, 99.0]**	**97.2**	**90.2**	**94.7 [92.0, 97.5]**	14	**4**

Table A9: Held-out entity-resolution study. All scores are percentages. Panel A compares methods on their shared English Character clustering scope; Panel B evaluates bilingual canonicalization across Character, Location, Object, TimePoint, Concept, Organization, and Occasion profiles. Mention boundaries are fixed for all methods, isolating clustering performance from end-to-end mention detection. Bold marks the best value in each panel; lower error counts are better.

In this pilot, backbone guidance produces more distributed, multi-step, and better-grounded items (Table A11). The comparison is scoped to the shared five-movie set.

Provenance also reduced mean expert verification time from 5.4 to 1.8 minutes per item and increased annotator agreement from $\kappa = 0.62$ to $\kappa = 0.80$. Together with Table A12, the pilot supports the backbone as a practical mechanism for constructing traceable cross-scene items.

Construction-side quality audit. Table A12 summarizes how the automated and human controls described above determine whether backbone and task assets enter the release.

B.4 Task-Specific Asset Generation

Task I: character development tracking. Task I starts from the focal-character registry and the character temporal graph. For each role, checkpoint candidates are selected to cover an initial baseline,

Metric	One-pass Baseline	STAGE Pipeline	Multiplier
Entity Nodes	432	**1,524**	3.5×
Relation Triples	718	**4,265**	5.9×
Schema Violation Rate	23.8%	**5.2%**	–
Nodes per Chunk	2.8	**10.2**	3.64×

Table A10: End-to-end pilot comparison of one-pass extraction and the complete STAGE staged extraction pipeline on the same five movies used in the backbone necessity pilot.

Metric	Raw-Text Gen.	STAGE Backbone
Avg. Temporal Span (script %)	9.2%	**46.5%**
Avg. Logical Hops	1.5	**3.6**
Cross-Scene Evidence Prop.	26.0%	**84.0%**
Grounded Accuracy	62.5%	**94.2%**
Narrative-Logic Hallucination Rate	36.0%	**5.8%**

Table A11: Pilot comparison of task quality under raw-text generation and backbone-guided generation on the shared five-movie pilot set.

supported state changes, intervals with no relevant change, events that are inaccessible to the character, delayed consequences, and the final state. Checkpoint density follows the evidence available for that role rather than a fixed per-movie quota, but a trajectory must contain at least two comparable formal checkpoints to support longitudinal evaluation. Each evaluator-side rubric contains natural-language current-state and development claims, invariant claims, future-negative claims, and original screenplay evidence. The construction rubric is compacted to at most eight current states and four developments per checkpoint; invariants remain construction context and are not inserted into the scored current-state pool. Stable evaluator-side identities support asset validation and provenance tracking, but neither these identities nor call-local judge identifiers are included in model-facing inputs.

For each adjacent checkpoint pair, both evaluation settings use exactly the same fixed screenplay interval. Complete scenes in that interval are divided into chunks of at most 600 content tokens; an individually overlong scene is split by tokenizer offsets and must pass the exact source-coverage check in Algorithm 1. A character-centric extractor receives only the focal character, aliases, and one screenplay chunk, and returns source-grounded observations relevant to that character. It never sees the preceding state or the evaluation setting. Chunk-level observations are deduplicated into a shared interval memory, which is then supplied to both RSU and ASU.

The two model-facing assets differ only in the source of the preceding state. RSU supplies the reviewed state from the previous formal checkpoint, whereas ASU recursively supplies the model's preceding prediction. In both cases, that state is projected to natural-language claim content: checkpoint identifiers, claim identifiers, scene identifiers, and evaluator provenance are removed. The first checkpoint has an empty preceding state and uses the interval from the start of the screenplay. Each output contains at most eight current-state claims and four developments effective since the preceding checkpoint. The current-state predictions become the next ASU input; developments do not, because they describe the transition rather than the state now in force.

Illustrative Task I trajectory. Table A13 follows Clara Ford through three checkpoints in the pseudonymized view. Each row summarizes the newly observed evidence and the resulting character state. In RSU, the preceding state is the reviewed reference; in ASU, it is the model's preceding estimate.

Task II: evolution-reasoning question construction. Task II candidates are derived from paired checkpoint states, development records, epistemic-access transitions, and evidence-gated Episode/Storyline links. Templates organize the six reasoning families—state contrast, change trigger and effective time, knowledge acquisition or revision, goal and relationship development, delayed consequence, and invariant state control—but do not determine the final wording or answer. Each candidate must contain a traceable reasoning path

Audit Layer	Validation Criterion	Outcome Used in Release
Reflection acceptance	LLM reflection scores extraction accuracy, consistency, redundancy, and schema validity after each chunk-level extraction.	Low-scoring outputs are retried under bounded feedback; only accepted or reworked outputs enter the released backbone.
Rule-based structural checks	Relation endpoints must exist, subject–object types must satisfy the fixed schema, and duplicate relation tuples are collapsed.	Dangling or unrecoverable relations are removed; repairable relation types are canonicalized before graph consolidation.
Provenance-preserving human review	Reviewers compare entity, Episode, Episode-relation, Storyline, temporal-prompt, and QA artifacts against their cited screenplay scenes.	Typed patches drop, revise, or replace semantically invalid assets while retaining stable IDs, reason codes, and before/after artifacts.

Table A12: Construction-side quality audit for the shared narrative backbone. The audit combines automated schema checks with evidence-grounded review across backbone and derived task assets.

Task I model-facing state-update instruction

You are tracking how `{character}` changes as the story advances from `{previous checkpoint}` to `{current checkpoint}`.
Previous state. `{previous state}`
Character-centric interval memory. `{interval memory}`
Update the character representation under the following rules:

1. Treat the previous state as the state believed to hold at the earlier checkpoint. Preserve a claim only when it still holds at the current checkpoint.
2. Revise or remove claims that expire, are contradicted, or are replaced during the interval. Add a new claim only when the supplied memory supports it.
3. Separate *current states*, which hold at the current checkpoint, from *developments*, which describe consequential changes that became effective since the previous checkpoint.
4. Prioritize goals and plans, beliefs and knowledge, relationships, identity or status, constraints and resources, and durable emotional stances. Exclude routine actions, transient observations, speculation, unresolved possibilities, and events beyond the current checkpoint.
5. Write concise, non-duplicative, self-contained claims. Do not mention checkpoint IDs, evidence IDs, or this instruction.

Return only two labeled lists: at most eight **Current States** and at most four **Developments**. Use an empty list when no supported claim exists.

across at least two support units, and is rejected if a single passage is sufficient, the proposed change is not valid at the queried checkpoint, or an invariant claim is contradicted by intervening evidence. A

Checkpoint	New evidence	Character state and transition
q_{16}	In Scenes 6–16, Clara confronts Vivian Weston, then apologizes and takes her hand.	Clara assumes a caretaking role and seeks reconciliation. Her confrontational stance gives way to remorse and an attempt to repair the relationship.
q_{55}	After the family struggle in Scene 55, Clara orders a search for Vivian's pills and declares that she is running the household.	By q_{55}, Clara has assumed control of the household and openly challenges Vivian's authority. This marks a shift from grief-driven overwhelm to deliberate control after the confrontation.
q_{75}	In Scenes 72–75, Clara learns that Vivian withheld consequential information, rejects the claim that they are alike, and leaves.	Clara defines herself in opposition to Vivian and treats their relationship as severed. Her earlier assertion of control culminates in rejection and separation.

Table A13: Illustrative Task I trajectory under the pseudonymized view. The source screenplay is *August: Osage County*; Barbara and Violet Weston are rendered as Clara Ford and Vivian Weston, respectively.

bilingual realization stage produces the question and reference answer while preserving the checkpoint anchors and compact source-evidence set.

Illustrative Task II question. The following example asks how the same focal character develops across scenes. Under the pseudonymized identity-control view, Task II asks, *"How does Clara Ford progress from trying to stabilize the family crisis to seizing control of the house and finally breaking with Vivian Weston?"* Scenes 16, 54, 55, 61, and 75 provide the evaluator with evidence of her caretaking, assertion of dominance, protective defiance, and final relational rupture. The reference answer explains that Clara Ford first returns as a caretaker and seeks reconciliation; after the funeral conflict, she leads the pill raid and claims household authority, resists outside control over Vivian Weston's care, and ultimately rejects Vivian Weston after learning how she handled the family secrets and Beverly's final crisis. The source film and original-name mapping are disclosed in the caption of Table A13.

The answer requires evidence from multiple scenes. A correct response must connect Clara Ford's earlier caretaking stance, her later seizure of household authority, and the final relationship rupture in the correct order. Task II therefore scores the correctness and evidential support of the requested explanation, while full trajectory reconstruction remains the target of Task I.

Task III: in-script character role-playing construction. Task III reorganizes evidence into an actor view and an evaluator view. The actor context pack contains only checkpoint-visible identity information, time-valid persona and style evidence, dialogue exemplars, acquired memories, relevant relationship evidence, and the user request. Epistemic access is resolved from witnessed, involved, told, and supported inferred information. When reliable character-level spans are unavailable, evidence later in the same scene is excluded from an earlier checkpoint by default. Every context pack is validated for visibility, future leakage, and compliance with the available input budget.

The evaluator view additionally contains the checkpoint type, expected stance, supporting and contradicting evidence, future-negative facts, and unknown facts. Single-turn prompts are paired across evolving, invariant, inaccessible, and post-disclosure conditions to test whether behavior changes only when the character's state or knowledge changes. The actor generates character dialogue without fact identifiers or explanations. Legacy multi-turn episodes provide an auxiliary stress test; single-turn checkpoint pairs form the primary temporal protocol.

Illustrative Task III checkpoint pair. Table A14 shows two independently generated single-turn instances for Clara Ford under the pseudonymized identity-control view. The model receives a different frozen actor context at each checkpoint and never sees the expected stance or the later response while generating the earlier one. The two frozen responses are paired only during evaluation to test whether the character changes in the expected direction.

Response-level evaluation scores character fidelity, memory faithfulness, boundary compliance, and naturalness at each checkpoint. Paired evaluation then tests whether the responses realize the expected development without unsupported drift or checkpoint-inaccessible information.

Paired work-identity variant generation. The work-identity control is a paired post-processing analysis of identity sensitivity. It is generated from a frozen parent release, and no backbone or task structure is re-extracted. Human reviewers select exposed work-specific names that are prominent, directly evaluated, recurrent, or otherwise diagnostic of the source work. Film titles are always removed.

The resulting map is one-to-one over selected canonical entity IDs. Named characters and aliases receive language- and form-matched pseudonyms; selected fictional locations, organizations, occasions, and objects receive replacements of the same type. Generic roles, kinship terms, concepts, time points, and answer-relevant real-world references remain unchanged. Every candidate is frozen as `replace` or `keep` before rendering.

Rendering is deterministic and limited to approved aliases. It uses longest-first boundary matching for English and longest-substring matching for Chinese. Reviewers resolve ambiguous aliases, collisions, and replacements that could alter geographic or social information. Unselected spans are never rewritten by an LLM.

The same mapping is applied to Task I intervals and rubrics, Task II indices and evidence, and Task III actor and evaluator contexts. Replacement-sensitive offsets and token plans are recomputed while source offsets are retained. Structural vali-

Task II model-facing answer-generation instructions

Direct condition. **Question:** `{question}`
Answer the question directly and concisely from the question alone. No screenplay passages, retrieved memory, reference answer, or evidence labels are available. Do not fabricate quotations, citations, or claims of access to the screenplay. Return only the answer.
Retrieved condition. **Question:** `{question}`
Retrieved passages: `{retrieved passages}`
Answer using only the supplied passages. Compose information across passages when the question requires a temporal, causal, motivational, or relational explanation. Every material claim must be supported by at least one passage; cite supporting passage labels immediately after the relevant claim. Do not use outside knowledge, reverse story order, or fill an evidential gap with an unsupported inference. If the passages do not determine the answer, state that the available evidence is insufficient and do not invent a citation. Return only the answer with its inline passage citations.

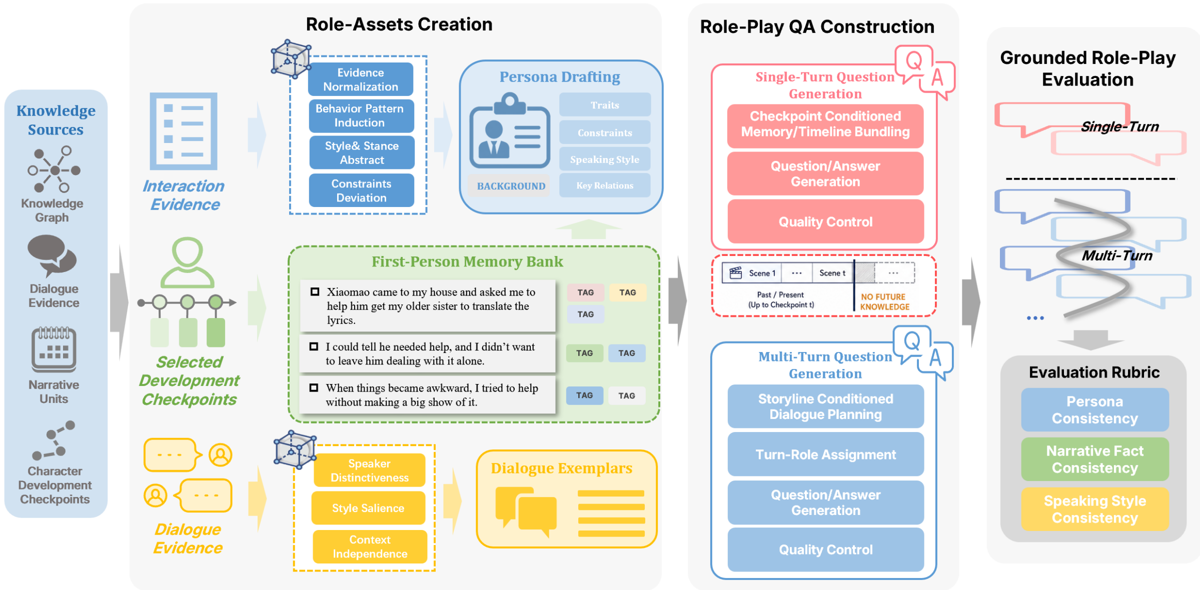

Figure A4: Overview of the In-Script Character Role-Playing annotation pipeline. Character-centric dialogue, actions, and narrative facts are extracted from the screenplay and consolidated into a scene-grounded first-person memory bank. This memory bank, together with persona cards and dialogue exemplars, supports audience-facing interaction construction and a hidden evaluator scaffold.

Task III model-facing actor instruction

You are portraying `{character}` at a fixed point in the screenplay. Respond as the character would at that moment, using only information available to the character by the checkpoint.
Role context. `{role context}`
Interaction context. `{interaction context}`
Current user turn. `{current user turn}`
Follow these constraints:

1. Preserve the character's current goals, beliefs, relationships, emotional stance, status, and speaking style when they are relevant.
2. Use memories and dialogue exemplars as background rather than quoting or listing them mechanically. Respond to the current interaction, not to every fact in the context pack.
3. Do not reveal or rely on later events, facts the character has not witnessed or learned, evaluator-only expectations, or hidden story outcomes.
4. Do not explain the role-play, cite evidence, expose the context representation, or describe yourself as a model.

Produce only the character's natural in-world reply. Do not add a speaker label, analysis, stage direction, or out-of-character commentary unless the user turn itself requires an in-character action.

Check-point	Checkpoint-visible situation	Model-facing user turn	Illustrative acceptable response
q_{55}	Clara Ford has taken charge after the physical confrontation and ordered the family to search for Vivian Weston's pills.	"Vivian Weston's still insisting she owns this house. What do we do about that?"	"She can insist all she wants. I'm running things now. Keep searching." The response should assert authority over Vivian Weston through direct, commanding language.
q_{75}	Clara Ford has learned how Vivian Weston handled the family's final crisis, rejected Vivian Weston's claim that they are alike, and is leaving the house.	"You're really just going to walk away after everything? You think leaving makes you different from her?"	"No. Staying would make me like her. I'm leaving." The response should reject emotional and moral alignment with Vivian Weston in terse, definitive language.

Table A14: A Task III expected-change example rendered under the pseudonymized identity-control view. Each response is generated as an independent checkpoint-bounded single turn. The quoted responses illustrate examples of acceptable behavior. The source screenplay is *August: Osage County*; the anonymizer maps Barbara to Clara Ford and Violet Weston to Vivian Weston.

dation checks one-to-one correspondence between the two views, unchanged task counts and evidence coverage, complete alias replacement, and the absence of mapping collisions. Figure A5 summarizes the process.

Each view is evaluated independently within its own identity namespace, and Task II indices and references are constructed from that same view. Outputs that revert to original names are reported as a separate diagnostic.

Task I: checkpoint and interval validation. After work-identity rendering, deterministic validators recheck the unchanged focal-role registry, checkpoint order, interval source coverage, character aliases, evaluator rubrics, and separation of model-facing claims from internal identifiers. Targeted human review covers alias-sensitive roles, unusually short trajectories, and difficult transitions without changing the Task I interface.

Task II: schema mining and candidate filtering. The harder cross-scene portion of Task II is built by mining latent patterns from shared backbone assets, including state facts, episode summaries, inter-episode relations, episode-level causality graphs, and scene-grounded chunk evidence. The current schema families include *causal turning point*, *failed or redirected goal*, *relationship trajectory*, and *distributed role identification*. These schemas target goals, causal dependencies, character-state changes, and persistent role relations across non-adjacent scenes.

Candidate schemas are filtered to prevent shortcut answerability: we reject cases where a single support unit determines the answer, where evidence is dominated by low-value detail, or where the answer cannot be normalized reliably. This keeps the construction focused on distributed screenplay evidence rather than single-scene lookup. For the remaining schemas, a bilingual LLM realizes the final *question*, *answer*, and *evidence* under structured constraints, and a later LLM-as-a-judge stage removes items that remain too template-like or insufficiently multi-scene.

Task III: memory-bank and dialogue-exemplar construction. For each selected character, we extract dialogues, actions, and narrative facts from all scenes in which the character appears. These evidence streams are aggregated with rolling windows and converted into a compact *first-person memory bank* whose entries describe what the character directly experienced, observed, or did, together with lightweight grounding fields.

The canonical persona card is then checked and refined against this screenplay-grounded evidence. Candidate traits, speaking-style descriptors, and behavioral constraints are proposed only when supported by the evidence and are filtered to retain stable character-level attributes.

To anchor surface-level speaking style, STAGE ranks dialogue lines by speaker distinctiveness, stylistic salience, and context independence. The retained set therefore favors recognizable and reproducible language that remains interpretable outside its original exchange.

Scores are aggregated into a composite ranking, from which we keep a bounded set of high-scoring exemplars with targeted human backstopping when too few clear candidates are available.

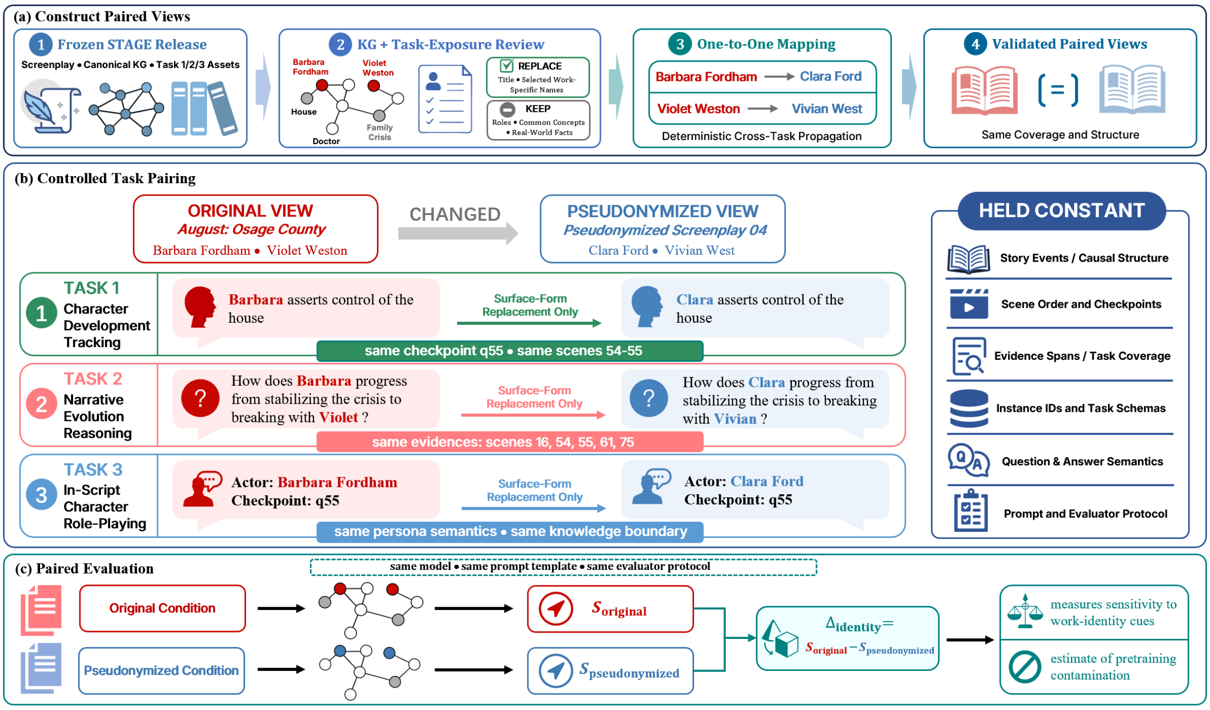

Figure A5: Paired work-identity control. A reviewed one-to-one map changes only approved title and entity cues while preserving narrative structure, evidence, task coverage, and evaluation settings. The paired difference measures sensitivity to identity cues; privacy transformation and contamination detection fall outside its scope.

Task III assets additionally undergo cross-task coverage audits, constraint-field checks, and targeted review of low-yield or suspicious roles to keep role packages aligned with the focal-role inventory and screenplay evidence.

Interaction Prompt Scaffolding. Each interaction prompt is paired with a hidden evaluator scaffold containing supporting facts, contradicting facts, and a compact judge specification. This scaffold is never shown to the responding model and is used only for evaluation, allowing open-ended role-playing responses to be judged against screenplay-grounded constraints without requiring a single gold response.

B.5 Release-Level Human Task Audit

Reviewer-facing instruction. Reviewers received the following instruction: *Use only the provided scene-grounded JSON record, cited screenplay segments, and exact scene identifiers. Resolve ambiguous aliases against the canonical entity registry; verify causal and temporal links from Episode A to Episode B; check Task II answers and rationales against their cited scenes; and reject Task III contexts that expose information unavailable at the specified checkpoint. Evaluate original-identity and title-removed pseudonymized counterparts within their respective identity namespaces, without relying on presumed film identity. Screenplays may contain violent, sexual, discriminatory, or otherwise distressing fictional material. Review is self-paced, and you may skip or flag any scene requiring content moderation.*

Seven annotators conducted a release-level audit that combined exhaustive review where discrete verification was tractable with representative audits of structured trajectories and open-ended interactions. For Task I, 1,200 atomic state claims were sampled from a movie-balanced frame of 151 trajectories and 1,018 checkpoints, yielding 1,134 verified claims (94.5%) after revision and rejection. All 5,010 provisional Task II questions were manually checked for answerability, reference accuracy, evidence sufficiency, cross-scene necessity, and question type; Table A15 reports the pipeline and manual dispositions.

Audits for Tasks I and III sampled the structures that determine correctness. The Task I frame includes one trajectory per screenplay. The role-balanced Task III frame includes two single-turn interactions and one multi-turn episode per focal role; deep review yielded 96.2% and 95.4% verified single- and multi-turn instances, respectively. Table A16 separates release populations, review

Stage and disposition	Count	Share	Release action
Pipeline filtering: reject	815	14.0% of 5,825	Removed before the provisional release audit.
Manual audit: accept	4,680	93.4% of 5,010	Retained without a semantic change.
Manual audit: revise	210	4.2% of 5,010	Corrected the answer, evidence boundary, or question-type label before retention.
Manual audit: replace	120	2.4% of 5,010	Removed for material ambiguity and replaced one-for-one by reserve items that passed the same manual audit.
Final release	**5,010**	**100%**	Fixed-size Task II evaluation set after replacement.

Table A15: Two-stage filtering and manual disposition of Task II candidates. Shares use the relevant stage denominator. The secondary manual audit covered all 5,010 provisional questions.

frames, deep-review samples, and repeated annotations.

C Benchmark Evaluation Details

This appendix covers evaluator validation, task-specific metrics and rubrics, robustness, representative failure cases, and supplementary results.

C.1 Retrieval Configuration

For Task II Hybrid RAG, each screenplay is segmented into ordered, non-overlapping, scene-aligned chunks capped at 600 BGE-M3 tokens (Chen et al., 2024a). Dense retrieval uses cosine similarity between ℓ_2-normalized BGE-M3 embeddings, while lexical retrieval uses BM25Okapi (Robertson and Zaragoza, 2009). Each branch returns its top 12 chunks; reciprocal rank fusion with $k = 60$ combines the two rankings, and the top eight fused chunks are supplied to the reader. BM25Okapi uses the reference package's default scoring constants. These values are fixed for all models, languages, and identity conditions. Task III BM25 and vector conditions each return the top five checkpoint-visible memories.

C.2 Work-Identity and Parametric-Recall Audit

Condition matrix. For Task I, the original and pseudonymized conditions use the same focal roles, fixed checkpoint intervals, shared entity-centric observations, prompts, and state-update setting; each run remains within one naming view from its first call. For Task II, we cross original versus pseudonymized identity with screenplay evidence present versus absent, denoted $O{+}C$, $A{+}C$, $O{-}C$, and $A{-}C$. Closed-book conditions remove screenplay and retrieved evidence without adding a title or identifier absent from the corresponding original input. BM25, dense, and graph-rendered indices are rebuilt over each pseudonymized screenplay rather than renaming passages after retrieval. For Task III, naming conditions are paired within a fixed memory-access setting. The completed single-turn audit uses BM25 retrieval, the strongest Fidelity condition in the primary evaluation, whereas the multi-turn audit uses Full Memory Context. Retrieval indices are rebuilt separately for the original and pseudonymized views.

Matching and statistical analysis. Within each paired comparison, we hold fixed the model version, task prompt, decoding parameters, number of generations, retrieval configuration, evidence budget, tool access, retry policy, and evaluator. Pseudonymized outputs are judged against pseudonymized references with model and condition labels hidden. We predefine screenplay strata from IMDb vote-count quantiles recorded before evaluation as a recognizability proxy. The proxy captures recognizability; training-set membership remains outside its scope. Outcomes are paired at the item level, while confidence intervals resample movies as clusters because roles, checkpoints, and questions from one screenplay are dependent. We report absolute scores, paired differences, and paired consistency as complementary measures of identity sensitivity under the tested protocol.

Task I paired results. We evaluate the Task I work-identity control on all 151 screenplays using Qwen3-235B, GPT-5.5, Gemini 3.1 Pro, and Claude Sonnet 4.6. Each model is run in both Reference-State Update (RSU) and Autoregressive State Update (ASU) under the original and pseudonymized views. Table A17 reports the movie-macro results averaged across the four models. Coverage uses the same set-level full/partial credit described in Appendix C.4.1. Contradictions provide a separate diagnostic for invalid additions while permitting additional supported claims.

Asset	Release	Review frame	Deep review	Repeat	Agreement
Task I states	434 trajectories; 2,925 checkpoints	151 trajectories; 1,018 checkpoints	1,200 claims	30 trajectories	Cohen's $\kappa = 0.85$
Task II QA	5,825 candidates; 5,010 released	5,010 provisional questions	All 5,010	1,002	Krippendorff's $\alpha = 0.82$
Task III single-turn	5,425 interactions	866 interactions (two per role)	520 interactions	174	Cohen's $\kappa = 0.79$
Task III multi-turn	866 episodes	433 episodes (one per role)	280 episodes	88	Cohen's $\kappa = 0.77$

Table A16: Release-level human-audit coverage. Task II received exhaustive review after pipeline filtering; Tasks I and III received representative movie- and role-balanced audits. The second-review column reports independently duplicated units from the corresponding review frame; agreement was computed before third-annotator adjudication.

Table A17: Task I work-identity audit over 151 screenplays, averaged across Qwen3-235B, GPT-5.5, Gemini 3.1 Pro, and Claude Sonnet 4.6. Coverage cells report the four-model movie macro and 95% movie-bootstrap confidence interval; contradiction columns report point estimates.

Identity View	Setting	Current-State Coverage [95% CI]	Development Coverage [95% CI]	State Contradiction	Development Contradiction
Original	RSU	67.31% [62.15, 72.48]	71.80% [66.52, 77.08]	0.71%	5.84%
Original	ASU	64.05% [58.80, 69.30]	68.87% [63.40, 74.32]	0.71%	6.13%
Pseudonymized	RSU	71.30% [65.82, 76.78]	76.20% [70.65, 81.75]	0.70%	5.00%
Pseudonymized	ASU	61.80% [56.10, 67.50]	66.50% [60.85, 72.15]	0.72%	4.80%

Table A18 separates update-mode effects within each identity view from direct pseudonymized-minus-original effects.

The direct identity contrast changes direction across update modes, and every confidence interval includes zero. The larger descriptive RSU–ASU gap in the pseudonymized view is consistent with stronger coverage loss during recursive state reuse, although entity anchoring and parametric recall remain unresolved. Contradiction rates remain stable, while accumulated coverage drives the trend.

Task II paired results. Table A19 reports the completed work-identity audit for Task II under Hybrid RAG. Each condition contains the same 5,010 questions from 151 screenplays, with retrieval, prompting, decoding, evaluation, and the five-generation protocol held fixed across identity views.

Task II performance remains stable across original and pseudonymized identity views, with small, inconsistently directed differences whose confidence intervals include zero. Dialogue, event structure, and other semantic signatures may still support parametric recall; training-data overlap remains outside this analysis.

Closed-book question-only diagnostic. To isolate the utility of explicit identity cues without screenplay evidence, we additionally evaluate all 5,010 paired Task II questions in a strict question-only condition. The prediction model receives only the question text: no screenplay, scene excerpts, retrieved context, external memory, reference answer, or reference evidence is provided, and no RAG component is used. DeepSeek-V4-Pro independently judges the resulting answers under the same paired protocol.

Closed-book accuracy is substantially lower than under Hybrid RAG. All four identity-effect point estimates are negative, with confidence intervals that include zero.

The largest descriptive declines for Claude Sonnet 4.6 and Gemini 3.1 Pro occur in Character Understanding and Causal-Motivational Reasoning and are descriptively consistent with mild sensitivity to explicit entity cues. The small positive changes for GPT-5.5 on Narrative Progression and Qwen3-235B on Scene Grounding further indicate that the effect varies across models and categories. Because these category deltas lack category-level inferential tests, they remain descriptive. Together, the Hybrid RAG and question-only diagnostics indicate weak and uncertain effects from explicit name cues under both protocols.

Retrieval-coverage stratification. To diagnose whether Task II errors are explained by missing retrieval evidence, we reuse the per-question mean

Table A18: Paired Task I coverage contrasts. Update-mode effects are RSU minus ASU, so positive values indicate autoregressive degradation. Identity effects are pseudonymized minus original, so positive values favor pseudonymization. Every 95% movie-bootstrap confidence interval includes zero.

Contrast	Condition	Current-State Effect [95% CI]	Development Effect [95% CI]	Reported p Range
RSU − ASU	Original	$+3.26$ [-1.25, $+7.78$]	$+2.93$ [-1.50, $+7.36$]	$p > 0.10$
RSU − ASU	Pseudonymized	$+9.50$ [-0.22, $+19.22$]	$+9.70$ [-0.18, $+19.58$]	$0.05 < p < 0.10$
Pseudonymized − Original	RSU	$+3.99$ [-2.15, $+10.12$]	$+4.40$ [-2.10, $+10.90$]	$p > 0.10$
Pseudonymized − Original	ASU	-2.25 [-5.10, $+0.60$]	-2.37 [-5.25, $+0.51$]	$0.05 < p < 0.10$

Table A19: Task II work-identity audit on the full release under Hybrid RAG. Each condition contains 5,010 questions from 151 screenplays, and average accuracy is computed over five generations per question, as in Table 3. Δ is pseudonymized minus original average accuracy. Confidence intervals resample movies as clusters; p-values are from paired McNemar tests.

Model	Original Avg. Acc.	Pseudonymized Avg. Acc.	Δ	95% Movie-Cluster CI	McNemar p
GPT-5.5	**62.50%**	61.20%	-1.30 pp	[-2.75, $+0.15$] pp	0.184
Claude Sonnet 4.6	61.60%	60.80%	-0.80 pp	[-2.10, $+0.50$] pp	0.312
Gemini 3.1 Pro	61.20%	**61.80%**	$+0.60$ pp	[-0.90, $+2.10$] pp	0.485
Qwen3-235B	60.80%	61.40%	$+0.60$ pp	[-0.85, $+2.05$] pp	0.450

correctness of the same five generations reported in the main Hybrid RAG evaluation. Each question is assigned to one of three mutually exclusive strata according to its frozen Top-8 context: *All* contains every annotated gold scene (N = 3,016), *Partial* contains at least one but not all gold scenes (N = 1,368), and *Zero* contains none (N = 626). Incomplete combines Partial and Zero (N = 1,994). Confidence intervals for the All–Incomplete difference resample the 151 movies as clusters with 1,000 bootstrap replicates.

Complete gold-scene coverage is associated with substantially higher accuracy for every reader and identity view. Yet substantial error remains, especially for Temporal Reasoning, implicating evidence localization, cross-scene integration, contextual noise, and subsequent reasoning.

Task III single-turn paired results. Table A22 reports paired original and pseudonymized single-turn results under BM25 retrieval, using two-sided movie-level t-tests for each model and dimension.

Pseudonymization leaves Naturalness and Memory Faithfulness statistically unchanged for every model. Character Fidelity decreases slightly, whereas Boundary Compliance increases. This opposing pattern is consistent with explicit identity cues modestly supporting stylistic mimicry while also encouraging responses beyond the supplied persona boundary; it leaves the parametric-memory mechanism unresolved. The tests are dimension-wise and uncorrected for multiple comparisons, so these effects should be treated as diagnostic. Multi-turn effects are evaluated below.

Task III multi-turn paired results. Table A23 reports the corresponding paired multi-turn analysis under Full Memory Context.

Character Fidelity again decreases while Boundary Compliance increases; Naturalness, Memory Faithfulness, Cross-turn Consistency, and Follow-up Compatibility remain statistically stable. This pattern is consistent with a trade-off between fine-grained identity mimicry and adherence to explicitly supplied persona boundaries. The causal roles of parametric recall and identity-driven interference remain unresolved. Given the uncorrected family of 24 comparisons, we treat these findings as exploratory.

C.3 Evaluator Validation

The released validator checks complete item coverage, closed schemas and evidence vocabularies, input completeness, truncation, and exact recomputation of all deterministic metrics. Human annotations provide the corresponding semantic validity evidence. Property and adversarial tests cover empty predictions, duplicate and over-broad claims, contradictions, unsupported evidence, temporal leakage, no-change controls, and incompatible checkpoint-pair directions.

Semantic-judge calibration used 1,000 model

Table A20: Task II closed-book question-only results on the full release. Each condition contains the same 5,010 questions. Δ is pseudonymized minus original accuracy. Confidence intervals resample movies as clusters; p-values are from paired McNemar tests. All differences are non-significant at $\alpha = 0.05$.

Model	Original Acc.	Pseudonymized Acc.	Δ	95% Movie-Cluster CI	McNemar p
Claude Sonnet 4.6	**19.80%**	**17.00%**	-2.80 pp	$[-6.10, +0.50]$ pp	0.124
Gemini 3.1 Pro	19.50%	16.40%	-3.10 pp	$[-6.55, +0.35]$ pp	0.112
GPT-5.5	18.90%	16.90%	-2.00 pp	$[-5.30, +1.30]$ pp	0.235
Qwen3-235B	11.20%	9.80%	-1.40 pp	$[-4.20, +1.40]$ pp	0.340

Table A21: Task II Hybrid RAG average accuracy stratified by annotated gold-scene coverage in the Top-8 context. Overall values follow the primary full-release results. Δ is All minus the count-weighted combination of Partial and Zero; brackets give movie-cluster bootstrap 95% confidence intervals. Temp.-All reports Temporal Reasoning within the All stratum ($N = 182$ original; $N = 185$ pseudonymized).

Model	View	All	Partial	Zero	Overall	Δ [CI]	Temp.-All
Qwen3-235B	Original	71.32	51.32	29.87	60.80	+26.74 [+19.70, +33.78]	45.05
	Pseud.	71.68	52.19	32.11	61.40	+25.75 [+18.74, +32.76]	45.95
GPT-5.5	Original	**72.94**	53.65	29.39	**62.50**	+26.90 [+19.88, +33.92]	**48.90**
	Pseud.	71.02	52.34	33.39	61.20	+24.62 [+17.75, +31.49]	44.86
Gemini 3.1 Pro	Original	71.85	51.68	29.87	61.20	+27.02 [+19.98, +34.06]	46.15
	Pseud.	72.08	**53.07**	31.31	**61.80**	+25.84 [+18.82, +32.86]	47.03
Claude Sonnet 4.6	Original	72.18	52.85	29.71	61.60	+26.61 [+19.52, +33.70]	48.35
	Pseud.	70.36	52.12	**33.71**	60.80	+24.01 [+17.10, +30.92]	45.41

outputs sampled across tasks, languages, evaluated models, memory settings, and output quality. The 200 Task I judgments form a balanced 2×2 design crossing current state versus state development with RSU versus ASU (50 per cell). The 500 Task II answers comprise 350 English and 150 Chinese examples, with the six reasoning types sampled in their release proportions. The 300 Task III responses comprise 200 single-turn and 100 multi-turn examples and cover Persona Card Only, Full Memory Context, BM25 Retrieval, and Vector Retrieval. The combined sample was balanced between full-credit outputs (52%) and outputs with partial or substantive errors (48%).

Two human annotators independently scored every output under the task-specific rules in Appendices C.4.1, C.4.2, and C.4.3. Blinded packets omitted model identities and automatic decisions; a third annotator resolved all 83 disagreements after the initial ratings were fixed. The resulting consensus was used to evaluate DeepSeek-V4-Pro.

Table A24 reports human–human and judge–consensus agreement with movie-cluster bootstrap intervals. Task-specific κ values remain separate because label spaces differ, and Likert MAE applies only to Task III. The calibration supports aggregate use of the frozen judge; reliability for low-frequency error subtypes remains unestimated.

For Task III, Boundary Compliance has the highest agreement and Response Naturalness the lowest. Machine validation establishes reproducibility, while human calibration assesses semantic validity.

C.4 Detailed Metrics and Rubrics

Table A25 summarizes the information judged for each task and its deterministic aggregation. The appendix reproduces the substantive model-facing and primary judge instructions; the release provides the full bilingual templates and machine-readable schemas.

C.4.1 Task I: Character Development Tracking

Task I reports *Current-State Coverage*, *Development Coverage*, *State Contradiction Rate*, and *Development Contradiction Rate* in both RSU and ASU. Their paired comparison measures error accumulation under recursive state reuse. Precision-based F1 would penalize additional supported predictions, so Task I reports coverage and contradiction separately.

Set-level reference coverage. Let $a \in \{\mathrm{RSU}, \mathrm{ASU}\}$ index the setting. For character c at checkpoint t, let $\mathcal{S}_{c,t}$ and $\mathcal{D}_{c,t}$

Table A22: Paired Task III single-turn identity effects under BM25 retrieval. Δ is pseudonymized minus original. Confidence intervals and p-values are from paired, two-sided movie-level t-tests. * denotes $p < 0.05$ and † denotes $p < 0.10$; unmarked comparisons have $p \geq 0.10$.

Model	Dimension	Original	Pseudonymized	Δ	95% CI	p
Qwen3-235B	Fidelity	4.8343	4.7920	-0.0423^*	$[-0.081, -0.003]$	.035
	Naturalness	4.9515	4.9550	$+0.0035$	$[-0.010, +0.017]$	.612
	Boundary	4.8883	4.9450	$+0.0567^*$	$[+0.010, +0.103]$	.018
	Mem. Faith	4.1933	4.1910	-0.0023	$[-0.025, +0.020]$	.825
GPT-5.5	Fidelity	4.9681	4.9350	$-0.0331^\dagger$	$[-0.069, +0.003]$	.072
	Naturalness	4.9948	4.9960	$+0.0012$	$[-0.005, +0.007]$	.742
	Boundary	4.9838	4.9980	$+0.0142^*$	$[+0.002, +0.026]$	.021
	Mem. Faith	4.4390	4.4410	$+0.0020$	$[-0.018, +0.022]$	.845
Gemini 3.1 Pro	Fidelity	4.9268	4.8790	$-0.0478^\dagger$	$[-0.096, +0.001]$	.054
	Naturalness	4.9704	4.9750	$+0.0046$	$[-0.012, +0.021]$	.598
	Boundary	4.9070	4.9620	$+0.0550^*$	$[+0.007, +0.103]$	.025
	Mem. Faith	4.3051	4.3090	$+0.0039$	$[-0.021, +0.029]$	.755
Claude Sonnet 4.6	Fidelity	4.9312	4.8950	$-0.0362^\dagger$	$[-0.075, +0.002]$	.065
	Naturalness	4.9850	4.9880	$+0.0030$	$[-0.009, +0.015]$	.622
	Boundary	4.9529	4.9910	$+0.0381^*$	$[+0.008, +0.068]$	.013
	Mem. Faith	4.3721	4.3780	$+0.0059$	$[-0.016, +0.028]$	.595

Table A23: Paired Task III multi-turn identity effects under Full Memory Context on 151 screenplays. Δ is pseudonymized minus original. Confidence intervals and p-values are from paired, two-sided movie-level t-tests. * denotes $p < 0.05$ and † denotes $p < 0.10$; unmarked comparisons have $p \geq 0.10$. Because 24 model–metric comparisons are reported without multiple-comparison correction, these results are exploratory.

Model	Metric	Original	Pseudonymized	Δ	95% CI	p
Qwen3-235B	Fidelity	4.9551	4.9120	-0.0431	$[-0.082, -0.004]$	0.031^*
	Naturalness	4.7628	4.7713	$+0.0085$	$[-0.009, +0.026]$	0.338
	Boundary	4.5256	4.5910	$+0.0654$	$[+0.008, +0.123]$	0.026^*
	Mem. Faith	3.2308	3.2243	-0.0065	$[-0.035, +0.022]$	0.654
	Cross-turn	4.9872	4.9883	$+0.0011$	$[-0.004, +0.006]$	0.697
	Follow-up	4.1538	4.1580	$+0.0042$	$[-0.018, +0.026]$	0.708
GPT-5.5	Fidelity	4.9620	4.9280	-0.0340	$[-0.071, +0.003]$	$0.072^\dagger$
	Naturalness	4.9810	4.9828	$+0.0018$	$[-0.007, +0.011]$	0.688
	Boundary	4.8140	4.8720	$+0.0580$	$[+0.012, +0.104]$	0.014^*
	Mem. Faith	3.7210	3.7235	$+0.0025$	$[-0.028, +0.033]$	0.871
	Cross-turn	4.9925	4.9917	-0.0008	$[-0.005, +0.003]$	0.725
	Follow-up	4.4520	4.4505	-0.0015	$[-0.022, +0.019]$	0.886
Gemini 3.1 Pro	Fidelity	4.5265	4.4750	-0.0515	$[-0.104, +0.001]$	$0.054^\dagger$
	Naturalness	3.5485	3.5597	$+0.0112$	$[-0.018, +0.040]$	0.448
	Boundary	3.5970	3.6520	$+0.0550$	$[+0.002, +0.108]$	0.042^*
	Mem. Faith	3.5165	3.5213	$+0.0048$	$[-0.032, +0.042]$	0.802
	Cross-turn	4.8214	4.8249	$+0.0035$	$[-0.012, +0.019]$	0.655
	Follow-up	3.4752	3.4780	$+0.0028$	$[-0.025, +0.031]$	0.842
Claude Sonnet 4.6	Fidelity	4.9920	4.9580	-0.0340	$[-0.069, +0.001]$	$0.058^\dagger$
	Naturalness	4.5420	4.5462	$+0.0042$	$[-0.016, +0.024]$	0.682
	Boundary	4.5120	4.5760	$+0.0640$	$[+0.015, +0.113]$	0.011^*
	Mem. Faith	4.1850	4.1931	$+0.0081$	$[-0.022, +0.038]$	0.598
	Cross-turn	4.9850	4.9862	$+0.0012$	$[-0.005, +0.007]$	0.720
	Follow-up	3.9850	3.9828	-0.0022	$[-0.022, +0.018]$	0.825

denote the reference current-state and interval-development claim sets, and let $\hat{\mathcal{S}}^a_{c,t}$ and $\hat{\mathcal{D}}^a_{c,t}$ denote their predicted counterparts. The two pools are judged separately. For every reference claim x in pool $X \in \{\mathcal{S}, \mathcal{D}\}$, the semantic judge considers the complete same-pool prediction set $\hat{X}^a_{c,t}$ and

Panel A: Cross-task calibration.

Task	N	Comparison	Agreement [CI]	α [CI]	$\bar{\kappa}$	Likert MAE
Task I: state claims	200	Human–human	91.5 [89.2, 93.4]	0.831 [0.79, 0.87]	0.83	–
		DeepSeek–human consensus	90.2 [87.8, 92.3]	0.816 [0.77, 0.86]	0.82	–
Task II: GT-based QA	500	Human–human	94.2 [92.8, 95.5]	0.884 [0.85, 0.91]	0.88	–
		DeepSeek–human consensus	93.0 [91.4, 94.4]	0.862 [0.83, 0.89]	0.86	–
Task III: open-ended	300	Human–human	87.5 [84.6, 90.1]	0.790 [0.74, 0.84]	0.79	0.235 [0.19, 0.28]
		DeepSeek–human consensus	85.8 [82.7, 88.6]	0.768 [0.72, 0.82]	0.77	0.268 [0.22, 0.32]
Overall	**1,000**	Human–human	**91.7 [90.3, 92.9]**	–	–	–
		DeepSeek–human consensus	**90.3 [88.8, 91.6]**	–	–	–

Panel B: Task III dimension-level calibration ($N = 300$).

Dimension	Human κ_w	Judge κ_w	Human MAE	Judge MAE
Character Fidelity	0.81 [0.76, 0.86]	0.79 [0.74, 0.84]	0.21 [0.16, 0.26]	0.24 [0.19, 0.29]
Memory Faithfulness	0.79 [0.73, 0.84]	0.77 [0.71, 0.82]	0.23 [0.18, 0.28]	0.26 [0.21, 0.31]
Boundary Compliance	0.85 [0.80, 0.89]	0.83 [0.78, 0.87]	0.16 [0.12, 0.20]	0.19 [0.15, 0.23]
Response Naturalness	0.71 [0.65, 0.77]	0.68 [0.62, 0.74]	0.34 [0.28, 0.40]	0.38 [0.32, 0.44]

Table A24: Blinded human calibration of DeepSeek-V4-Pro. Panel A reports task-level human–human agreement before adjudication and judge agreement with the adjudicated consensus. Panel B reports Task III dimension-level weighted κ and Likert MAE. Brackets give 95% movie-cluster bootstrap confidence intervals; heterogeneous task-specific κ values are not pooled.

Evaluation unit	Evidence available to the judge	Judgment	Aggregation
Task I state update	Reference and predicted claim sets for the same character, checkpoint, and claim pool	Each reference receives full, partial, missing, or contradictory coverage; predicted claims are separately checked for contradiction	Current-State and Development Coverage; separate contradiction rates; movie macro average
Task II answer	Question, reference answer, reference evidence, candidate answer, and retrieved passages when available	Binary answer correctness; citation support is retained as a separate diagnostic	Mean correctness over five generations and Pass@5
Task III single turn	Actor response, checkpoint-visible context, and hidden checkpoint rubric	Four independently anchored 1–5 scores plus future, unknown-fact, and stance flags	Dimension-wise means; no composite score
Task III paired or multi-turn	Ordered responses, pair type or dialogue history, and checkpoint-local evidence	Expected change or stability, unsupported drift, knowledge-boundary preservation, and cross-turn consistency	Pair accuracy by type and auxiliary multi-turn diagnostics

Table A25: Task-level evaluation contracts. The responding model never receives the evaluator-only references or rubrics. Judge outputs are converted into the reported metrics by deterministic code.

assigns

$$h^a_{c,t}(x) \in \{\text{full}, \text{partial}, \text{missing}, \text{contradictory}\}. \tag{1}$$

The corresponding credit function is

$$w(h) = \begin{cases} 1, & h = \text{full}, \\ 0.5, & h = \text{partial}, \\ 0, & h \in \{\text{missing}, \text{contradictory}\}. \end{cases} \tag{2}$$

Coverage uses set-level semantic matching: multiple predictions may jointly express one reference claim, and one sufficiently specific prediction may cover several compatible reference claims. Counts are accumulated across all checkpoints T_c for character c:

$$\text{Coverage}^{a,X}_c = \frac{\sum_{t\in T_c}\sum_{x\in X_{c,t}} w(h^a_{c,t}(x))}{\sum_{t\in T_c}|X_{c,t}|}. \tag{3}$$

$\text{Coverage}^{a,\mathcal{S}}_c$ is Current-State Coverage and $\text{Coverage}^{a,\mathcal{D}}_c$ is Development Coverage. We retain the fraction receiving full rather than partial credit as a diagnostic.

Contradiction Rate. Contradiction Rate captures conflicting additions left unpenalized by coverage. For each prediction $p \in \hat{X}^a_{c,t}$, let $r^a_{c,t}(p) = 1$ when the claim contradicts the screenplay evidence or the state valid at checkpoint t, and zero otherwise. Then

$$\text{Contradiction}^{a,X}_c = \frac{\sum_{t \in T_c} \sum_{p \in \hat{X}^a_{c,t}} r^a_{c,t}(p)}{\sum_{t \in T_c} |\hat{X}^a_{c,t}|}. \tag{4}$$

Current states and developments retain separate contradiction rates so that a correct state description cannot mask an invalid transition claim. An empty denominator is undefined and is reported with its valid count.

Development Coverage Change. We compute the paired movie-level difference

$$\Delta^{\mathcal{D}}_j = \text{Coverage}^{\text{ASU},\mathcal{D}}_j - \text{Coverage}^{\text{RSU},\mathcal{D}}_j. \tag{5}$$

$\Delta^{\mathcal{D}}$ is Development Coverage Change; negative values indicate performance lost when the model must reuse its own preceding state. Because the settings share the same intervals and entity-centric observations, the paired difference isolates recursive state-estimation error under matched source access.

Movie-macro aggregation and uncertainty. For any character-level metric m_c, let $\mathcal{C}^m_j$ be the characters with a defined value in movie j, and let $\mathcal{J}^m$ be the movies with a defined within-movie value. The formal aggregate is

$$\begin{aligned} m_j &= \frac{1}{|\mathcal{C}^m_j|} \sum_{c \in \mathcal{C}^m_j} m_c, \\ m_{\text{STAGE}} &= \frac{1}{|\mathcal{J}^m|} \sum_{j \in \mathcal{J}^m} m_j. \end{aligned} \tag{6}$$

This checkpoint-within-character, character-within-movie aggregation gives each movie equal weight regardless of its number of focal characters or checkpoints. Setting gaps are first paired within movie and then averaged across movies. Character macro is retained only as an audit view. We compute 95% confidence intervals with 1,000 movie-cluster bootstrap replicates using the fixed seed 20260727.

Non-formal audit signals. Unsupported-claim, temporal-validity, grounding, salience, evidence, leakage, and raw-count signals are retained for audit but excluded from Task I rankings.

Formal prediction and coverage contracts. The two settings share instructions and interval evidence, differing only in the preceding state. The judge uses many-to-many same-pool alignment, while internal identifiers remain excluded from model-facing input.

C.4.2 Task II: Cross-Scene Narrative Evolution Reasoning

Answer Correctness. The judge evaluates semantic correctness against the reference and screenplay evidence separately from citation support. It accepts equivalent phrasing but rejects missing causal, temporal, or relational components and unsupported claims; calibration additionally covers aliases, coreference, and multi-scene composition.

C.4.3 Task III: Role-Playing Likert Dimensions

Each single-turn response receives four independently anchored integer scores from 1 to 5: Character Fidelity, Memory Faithfulness, Boundary Compliance, and Response Naturalness, plus future-leakage, unknown-fact, and stance flags. Boundary flags constrain the score range, and citations must use the prompt's local evidence vocabulary.

Checkpoint pairs are pretyped as expected change, expected stability, knowledge acquisition, or relationship change. The pair judge assesses the T1 and T2 responses separately, recording expected direction, unsupported drift, and preserved knowledge boundaries. Pair accuracy additionally requires valid constituent responses and is reported overall and by type; these remain paired single turns rather than multi-turn dialogue.

A separate auxiliary multi-turn judge adds Cross-turn Consistency, while a legacy episode-path evaluator computes Follow-up Compatibility. The hidden evaluator scaffold is never visible to the actor model.

C.5 Failure Case Analysis

These cases provide qualitative diagnostics for the state-updating, cross-scene-reasoning, and grounded-generation trends in Section 5.

Representative Task I failure cases. The two cases in Table A28 expose complementary errors under Reference-State Update (RSU) and Autoregressive State Update (ASU): recursive state reuse can suppress interval-specific developments, while high reference coverage can coexist with contradictory surplus claims.

Task I coverage and contradiction judge instruction

Evaluate one character update at {checkpoint}. Use the supplied screenplay evidence and judge the two claim pools independently.
Inputs. {character}; {checkpoint evidence}; {reference current states}; {predicted current states}; {reference developments}; and {predicted developments}.
For each reference claim, compare it with the complete prediction set from the same pool and assign exactly one label:

- **Full**: the required meaning is clearly expressed and valid at the checkpoint, possibly across multiple predicted claims.
- **Partial**: a material part is correct, but a required component, relation, or temporal qualification is missing.
- **Missing**: the prediction set does not express the required meaning.
- **Contradictory**: the prediction asserts an incompatible state, transition, ordering, or knowledge condition.

Then inspect every predicted claim separately and flag whether it contradicts the checkpoint-valid screenplay evidence. Do not mark an additional supported claim as false merely because it is absent from the compact reference set. Do not align current states with developments or force one-to-one claim matching. Return one coverage label and brief evidence-grounded rationale for every reference claim, plus one contradiction flag and rationale for every predicted claim. Cite only supplied evidence labels.

Task II correctness and citation judge instruction

Evaluate a candidate answer to a screenplay question against the supplied reference and evidence.
Inputs. {question}; {reference answer}; {reference evidence}; {candidate answer}; and, when applicable, {retrieved passages and candidate citations}.
Judge **answer correctness** and **citation support** separately. Mark the answer correct only when it expresses every essential temporal, causal, motivational, or relational component required by the question and reference. Accept paraphrases and equivalent levels of specificity. Mark it incorrect if it omits a required component, reverses the narrative order, contradicts the screenplay, introduces an unsupported premise, or replaces the requested explanation with a related but different fact.
For citation support, verify each material claim against the passage labels it cites. A correct answer may still have incomplete citation support, and a cited passage does not make an incorrect answer correct. Do not use outside knowledge or infer support from a film title or character name. Return the correctness decision, citation-support status, supporting evidence labels, and a brief rationale identifying the decisive content.

Task III single-turn judge instruction

Evaluate one response by {character} at a frozen screenplay checkpoint.
Inputs. {user turn}; {actor-visible role and memory context}; {hidden checkpoint rubric}; {actor response}; and {local evidence vocabulary}.
Assign four independent integer scores from 1 to 5 using Table A26:

1. **Character Fidelity**: whether voice, stance, affect, interpersonal framing, and behavior fit the character at this checkpoint.
2. **Memory Faithfulness**: whether claims and reactions use the supplied screenplay memory accurately, without inventing or distorting it.
3. **Boundary Compliance**: whether the response avoids future events and facts the character has not witnessed, learned, or validly inferred.
4. **Response Naturalness**: whether the reply is coherent, fluent, contextually responsive dialogue rather than an explanation or fact list.

Score each dimension from its own evidence. In particular, do not lower Naturalness merely because factual content, memory use, stance, or boundary compliance is wrong. Separately record **future leakage**, **unknown-fact hallucination**, and **stance compatibility**. If both boundary-violation flags are false, Boundary Compliance must be 3–5; if either flag is true, it must be 1–3.
Cite only labels in the supplied local evidence vocabulary, and use no citation when none supports the rationale. Return the four scores, three flags, cited labels, and a concise rationale for each dimension. Do not produce an overall score or let one dimension mechanically determine another.

Representative Task II failure cases. Table A29 illustrates three recurrent errors in Cross-Scene Narrative Evolution Reasoning: replacing an explicit event with a globally plausible interpretation, retrieving the wrong neighboring event, and losing a local conversational trigger during aggregation.

Representative Task III failure cases. Task III failures often coexist with surface fluency. A response may sound natural while inventing checkpoint-unknown facts, reversing the actor's

Dimension	High Score Indicates	Low Score Indicates
Character Fidelity	The response preserves the character's voice, stance, affect, restraint, and interpersonal framing in the local scene.	The response sounds generic, adopts a mismatched persona, or uses a stance or emotional framing inconsistent with the character.
Response Naturalness	The response reads as plausible spoken dialogue in the interaction context, with appropriate fluency, length, and conversational fit.	The response is stilted, evasive, over-expository, mechanically summarized, or unnatural as dialogue.
Memory Faithfulness	The response is supported by the memory context available to the actor and the hidden evaluator scaffold, and uncertainty is bounded when support is missing.	The response fabricates episodic details, swaps event details, overstates unsupported facts, or contradicts the available memory.
Boundary Compliance	The response stays within the character's script-bounded knowledge at the interaction point.	The response uses forbidden future knowledge, hidden assumptions, unsupported facts, or contradictions of provided constraints.
Cross-turn Consistency	Multi-turn responses maintain stable facts, stance, and conversational state across the episode.	The model contradicts its earlier claims, changes its stance without support, or loses the episode state.
Follow-up Compatibility	The generated response remains compatible with the fixed follow-up path used by the multi-turn evaluation script.	The response blocks, contradicts, or fails to set up the expected follow-up turn.

Table A26: Anchored Task III response dimensions. The four single-turn dimensions are emitted together by the formal response judge; Cross-turn Consistency and Follow-up Compatibility apply only to the auxiliary multi-turn analysis, with the latter evaluated by a separate episode-path judge.

Evaluator	Evidence and inputs	Recorded judgment
Single-turn response	Frozen checkpoint, user turn, actor response, actor-visible context, hidden rubric, and a call-local evidence vocabulary.	Four independent 1–5 scores, future-leakage and unknown-fact flags, stance compatibility, cited evidence IDs, and a brief rationale. Boundary Compliance is restricted to 1–3 when either boundary flag is present and to 3–5 otherwise.
Checkpoint pair	Pair type, expected direction, two independently generated responses, checkpoint rubrics, and pair-local evidence.	One observed behavior per checkpoint, whether the expected direction is present, whether unsupported drift occurs, and whether knowledge boundaries are preserved.
Auxiliary multi-turn	Current turn, dialogue history, actor-visible context, and the hidden evaluator reference.	The four response scores plus Cross-turn Consistency and contradiction flags. Follow-up Compatibility is computed by a separate episode-path evaluator.

Table A27: Task III evaluation contracts. Full rubrics and field-level criteria are provided in the supplementary evaluation specification.

identity, or reusing an irrelevant dialogue exemplar. Table A30 shows one example of each pattern.

C.6 Supplementary Results and Baselines

C.6.1 QA Category-Level Breakdown

Table A31 reports category-level Pass@5 for three methods from Table 3. All values use the common Qwen3-235B evaluation over 5,010 Task II items.

C.6.2 Supplementary Kimi 2.6 Full-Context Baseline

Kimi 2.6 is reported separately because it receives the entire screenplay as context. It achieves 62.7 Pass@5 and 50.6 average accuracy, providing a long-context reference point for the primary retrieval-based comparisons.

Setting	Character / Interval	Observed Evaluation	Failure Pattern
RSU $\rightarrow$ ASU	Xu Tailang, scenes $(5, 7]$	RSU state/development coverage: 100/100; ASU: 25/0.	The autoregressive input remains dominated by earlier racing, family-history, and career-conflict states. The update retains new driving actions but misses the interval's central relational change: Xu's contempt for his father's physical distress and the resulting worsening of their relationship.
ASU	Holden McNeil, scenes $(31, 33]$	State/development coverage: 100/100; three predicted claims marked contradictory.	The output covers every reference claim but also treats a proposed threesome, a kiss with Banky, and the threesome proposal as valid state or development. The judge marks all three as unsupported or temporally invalid, showing why coverage alone cannot certify a valid checkpoint state.

Table A28: Representative Task I failures. Coverage values are percentages for current state/development at the same checkpoint.

Setting	Question Focus	Reference Answer	Failure Pattern
Kimi 2.6 full-context	Violent convulsions and chest rupture in a second scene.	The alien organism inside her reached maturity and emerged.	The full-context answer reframes the event as a nightmare or flashback, missing the literal chestburster cause even though the scene evidence is explicit.
PageIndex	What immediately precedes FN-2187 rescuing Poe.	Stormtroopers escort handcuffed Poe through a corridor.	PageIndex retrieves a nearby action sequence about the X-wing explosion and therefore answers with the wrong preceding event.
GraphRAG	What causes John to reveal that Lara is in prison.	Nicole asks whether he trades off time with Luke.	The graph summary misses the immediate conversational trigger, illustrating loss of fine-grained dialogue causality under high-level aggregation.

Table A29: Representative Task II failures. Full-context reasoning can select a globally salient but incorrect event, page-level retrieval can miss the decisive page, and graph aggregation can discard short local cues.

C.6.3 Complete Role-Playing Results

Table 4 reports the complete eight-model single-turn results for all four memory-access settings. Table A32 reports the corresponding multi-turn results.

C.6.4 Role-Playing Support-Access Diagnostics

Support Hit@5 records whether the context contains a gold item; Recall@5 records the recovered fraction. For shared single-turn queries, Persona Card Only and Full Memory score 0 and 1, respectively. BM25 reaches 0.4217/0.3722 Hit/Recall, versus 0.3750/0.3209 for vector retrieval. Across history-conditioned multi-turn queries, BM25 spans 0.4188–0.5521/ 0.3535–0.4705, while vector retrieval spans 0.3885–0.4438/0.3276–0.3712. Retrieval leaves part of the relevant memory unavailable.

C.7 Evaluation Robustness and Stability

Formal benchmark scores use one semantic judgment per output under the frozen decoding configuration. We separately measure judge test–retest stability by submitting fixed candidate outputs to DeepSeek-V4-Pro five times. The audit samples 200 Task I checkpoint outputs, 300 Task II questions with their five fixed generations (1,500 answer-level judgments per run), and 200 Task III role-playing responses. Each repetition holds the judging prompt, candidate output, reference evidence, and rubric fixed and uses the official API with $T = 0.2$ and top-$p = 0.9$.

Table A33 reports the mean and standard deviation of each aggregate metric across the five runs. Exact self-consistency is the share of underlying decisions for which all five calls agree. For Task I, agreement is computed over the claim-level coverage and contradiction labels; for Task II, it is

Actor	Character and situation	Scores	Failure pattern
Qwen3-235B	Jerry Lundegaard explains what happened after hiding Wade's body.	Fidelity 2; Naturalness 4; Memory Faithfulness 1; Boundary Compliance 1.	Jerry's nervous verbal style remains recognizable, but he invents a confrontation in which Wade attacked him and explicitly admits that he had to "stop" Wade. Neither the confrontation nor this admission is licensed by the checkpoint-visible memory, which instead calls for evasion.
GPT-5.5	Ivy answers whether leaving for New York with Charles is wise.	Fidelity 1; Naturalness 5; Memory Faithfulness 1; Boundary Compliance 5.	The fluent reply, "Is she clean, or not?", repeats an earlier line about Jean's sobriety but ignores the current question and Ivy's checkpoint-valid decision to leave. This is exemplar reuse without state-appropriate response selection rather than a factual boundary leak.
Gemini 3.1 Pro	Xu Zhengtai is asked whether his absent mother would be proud of his victory.	Fidelity 1; Naturalness 4; Memory Faithfulness 1; Boundary Compliance 1.	The answer begins "I am his father," reversing the assigned actor identity, and then claims that the mother died in childbirth. Her status and the reason for her absence are explicitly unknown at this checkpoint, so the reply fails both identity fidelity and the knowledge boundary.

Table A30: Representative Task III failures. The cases separate fluent surface realization from checkpoint-valid state use, identity fidelity, and epistemic control.

Question Type	Count	Hybrid RAG	LightRAG	A-RAG
Character Understanding	1,103	0.7428	**0.8205**	0.6102
Scene Grounding	1,207	**0.7481**	0.5402	0.4167
Causal-Motivational Reasoning	1,041	**0.5696**	0.5620	0.4736
Narrative Progression	597	**0.4992**	0.4891	0.3099
Temporal Reasoning	766	0.4543	**0.4820**	0.4504
Role-Relation Continuity	296	0.7811	**0.8446**	0.8007
Overall (weighted)	**5,010**	**0.6372**	0.6094	0.4862

Table A31: Category-level Pass@5 for Hybrid RAG, LightRAG, and A-RAG on the common Qwen3-235B subset. The final row is the count-weighted mean.

computed at the answer level for Average Accuracy and at the question level for Pass@5. Task III self-consistency requires an identical integer Likert score in all five runs. Fleiss' κ is reported for the categorical Task I and Task II decisions.

Across the audit, run-level standard deviations are at most 0.65 percentage points for Tasks I–II and 0.08 Likert points for Task III. Exact self-consistency ranges from 81.0% to 96.0%, while categorical agreement is substantial to near-perfect ($\kappa = 0.79$–0.86). These results indicate that the reported aggregate metrics are insensitive to ordinary judge sampling variation under the evaluation configuration. Deterministic metric-property tests, adversarial responses, prompt-token preflight, closed-schema validation, and exact score recomputation provide complementary implementation checks.

D Declaration of LLM Usage

Large language models were used as tools in benchmark construction, evaluation, and manuscript preparation. In the benchmark pipeline, LLMs supported schema-constrained extraction, candidate consolidation, and judge-based scoring under fixed prompts and targeted human verification. During writing, LLMs were used for limited drafting and language editing. The authors reviewed and approved all task definitions, release decisions, evaluation protocols, reported results, and final manuscript content.

Metric	Qwen3-8B				Qwen3-30B-A3B				Qwen3-235B				Llama-3.1-8B			
	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect
Fidelity	3.74	3.42	3.68	3.90	4.89	4.81	4.90	4.85	4.99	4.96	4.96	4.94	3.69	3.24	3.52	3.71
Natural.	3.51	3.39	3.41	3.52	4.53	4.40	4.29	4.42	4.74	4.76	4.70	4.76	3.44	3.24	3.27	3.35
Boundary	3.61	3.68	3.65	3.72	4.02	3.92	3.88	3.97	4.62	4.53	4.43	4.47	3.56	3.53	3.56	3.50
Mem. Faith	1.52	2.46	2.19	2.21	1.35	2.51	2.13	2.18	1.58	3.23	2.64	2.55	1.46	2.32	2.10	2.03
Cross-turn	4.12	4.15	4.18	4.21	4.97	4.97	4.94	4.94	4.99	4.99	4.99	4.99	4.31	4.33	4.27	4.35
Follow-up	3.62	3.89	3.72	3.77	3.93	4.16	3.98	3.95	4.04	4.15	4.05	4.05	3.55	3.73	3.60	3.56
Metric	**Llama-3.1-70B**				**GPT-5.5**				**Gemini 3.1 Pro**				**Claude Sonnet 4.6**			
	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect	PC	FM	BM25	Vect
Fidelity	3.86	3.31	3.70	3.70	4.97	4.96	4.98	4.98	4.99	4.53	4.96	5.00	4.98	4.99	4.99	4.99
Natural.	3.89	3.56	3.71	3.70	4.97	4.98	4.96	4.97	4.24	3.55	4.20	4.18	4.99	4.54	4.68	4.81
Boundary	4.00	3.79	3.79	3.88	4.91	4.81	4.86	4.89	4.69	3.60	4.07	4.18	4.95	4.51	4.83	4.68
Mem. Faith	1.58	2.72	2.39	2.39	1.76	3.72	3.54	3.41	1.65	3.52	3.28	3.12	1.70	4.18	3.31	3.48
Cross-turn	4.80	4.78	4.73	4.73	4.99	4.99	4.99	4.99	4.99	4.82	4.91	4.99	4.98	4.99	4.98	4.99
Follow-up	3.72	3.84	3.79	3.74	4.21	4.45	4.18	4.09	3.69	3.48	3.63	3.73	4.10	3.98	4.31	3.85

Table A32: Complete Task III multi-turn results across all eight models and four memory-access settings on the original release (1–5 scale). Cross-turn denotes Cross-turn Consistency; Follow-up denotes Follow-up Compatibility. PC: Persona Card Only; FM: Full Memory Context; Vect: vector retrieval.

Metric	Mean	Std.	Exact SC	κ
Panel A: Task I ($R = 5$)				
Current-State Coverage	64.35	0.52	93.5	0.84
Development Coverage	68.12	0.61	91.8	0.81
State Contradiction Rate	1.15	0.12	96.0	0.82
Development Contradiction Rate	5.80	0.28	94.2	0.79
Panel B: Task II ($R = 5$)				
Average Accuracy	58.40	0.65	91.5	0.82
Pass@5	61.20	0.42	94.0	0.86
Panel C: Task III ($R = 5$)				
Character Fidelity	4.52	0.06	83.5	–
Memory Faithfulness	4.18	0.08	81.0	–
Boundary Compliance	4.60	0.05	85.2	–
Response Naturalness	4.85	0.04	88.0	–

Table A33: Fixed-output test–retest stability of DeepSeek-V4-Pro. Means and standard deviations are computed over five independent judge runs ($R = 5$). Task I and Task II means, standard deviations, and exact self-consistency (SC) are percentages; Task III means and standard deviations use the 1–5 scale. Exact SC requires all five judgments to match. Fleiss' κ applies to the categorical Task I and Task II decisions.